

Topology-Aware Decentralized Communication Control for Dynamic Clustered FANETs

Himanshu Kumar and Atri Mukhopadhyay

Abstract—Flying ad hoc networks (FANETs) require communication-control mechanisms that can remain effective under rapid topology changes, uneven traffic concentration, spatially varying interference, and local failures of coordination-critical nodes. In such settings, a single network-wide medium-access decision is often too coarse to capture the different communication demands experienced across the swarm. This paper addresses that limitation by formulating FANET communication control at the level of dynamic clusters, where cluster-heads act as decentralized decision makers and interact over a time-varying cluster interference graph. Within this framework, each cluster-head jointly determines its local access behavior and how burst time should be divided between intra-cluster service and inter-cluster coordination. The resulting formulation integrates dynamic clustering, graph-aware decentralized control, adaptive burst allocation, and local fault-tolerant leader handover into a unified architecture. To solve this problem, the paper develops a graph-aware multi-agent reinforcement-learning controller with recurrent memory and dueling action evaluation, and compares it against a structured centralized baseline that preserves the same cluster-level decision semantics. The study is designed to evaluate not only throughput, delay, drops, collisions, and fairness, but also topology sensitivity, burst-allocation behavior, clustering dynamics, leader-failure recovery, robustness to train–test mismatch and imperfect graph information, architectural ablations, and deployment-facing complexity. Overall, the paper presents a mathematically grounded and experimentally disciplined framework for decentralized, topology-sensitive, and robust communication management in dynamic clustered FANETs.

Index Terms—Flying ad hoc networks, decentralized control, dynamic clustering, graph neural networks, multi-agent reinforcement learning, burst scheduling, fault tolerance, wireless resource allocation.

I. INTRODUCTION

FLYING ad hoc networks (FANETs) are increasingly expected to support sensing, relaying, distributed coordination, and mission-critical information exchange under rapid topology variation, intermittent connectivity, and highly nonuniform traffic demand. Unlike many terrestrial wireless systems, FANETs operate in a regime where mobility, interference, queue evolution, and coordination overhead change on comparable time scales. As a result, the communication-control problem cannot be reduced reliably to a single network-wide medium access decision. Different regions of

the swarm may simultaneously experience different backlog pressure, relay burden, coordination demand, and interference exposure, even within the same mission epoch. This creates a need for control policies that are explicitly decentralized, topology-sensitive, and robust to local disruptions.

A. Background and Motivation

A large body of work on wireless scheduling and resource allocation has shown that scalable control in interference-coupled networks depends critically on locality, information structure, and the ability to react to spatial heterogeneity. These requirements become more severe in FANETs because the underlying communication graph evolves continuously with UAV mobility, while the network must still sustain both local service and cross-network coordination. Consequently, a practically useful FANET controller must satisfy two conditions simultaneously. First, it must operate from partial, spatially localized information rather than assuming persistent access to a consistent global state. Second, it must remain sensitive to the interdependence induced by neighboring clusters, shared spectrum, relaying demand, and coordination traffic.

To address these requirements, this paper adopts a dynamic clustered view of the FANET. Let the network contain n UAVs, and let $c(t)$ denote the number of active clusters at decision instant t . The swarm is partitioned into a time-varying collection of clusters,

$$\mathcal{U}(t) = \bigcup_{k=1}^{c(t)} C_k(t), \quad (1)$$

where each cluster $C_k(t)$ is represented by a cluster-head $L_k(t) \in C_k(t)$. The cluster-head becomes the natural decentralized decision unit because it aggregates local service state while also participating in inter-cluster coordination. This representation preserves a deployment-facing interpretation: subordinate UAVs need not access the full network state, whereas cluster-heads can act from cluster summaries and neighborhood information over a dynamic cluster interference graph.

B. Why Global MAC Selection Is Insufficient

A network-wide medium access control (MAC) decision assumes that the FANET can be described by a sufficiently homogeneous communication condition at each decision instant. In the present setting, that assumption is structurally weak. Spatially separated clusters may experience markedly different

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queue occupancy, interference intensity, mobility stability, and relay demand, even while operating in the same network episode. A single global MAC mode therefore suppresses the very heterogeneity that should drive adaptation.

More fundamentally, the control objective here is not only to select a nominal local access mode. Each cluster must also decide how much of its limited burst budget should be devoted to local service within the cluster and how much should be reserved for coordination with neighboring clusters. A global MAC-selection abstraction cannot express this tradeoff faithfully, because it collapses spatially localized service needs and graph-coupled coordination needs into one broadcast decision. Such a formulation weakens both the mathematical meaning of decentralization and the physical interpretation of resource allocation in clustered FANETs.

C. Why Decentralized Cluster-Head Control Is Needed

The proposed architecture therefore treats cluster-heads as the decentralized agents. Each cluster-head $L_k(t)$ governs a local cluster $C_k(t)$ while interacting with neighboring cluster-heads over a dynamic cluster interference graph

$$G_t = (V_t, E_t), \quad V_t = \{1, 2, \dots, c(t)\}, \quad (2)$$

where vertices correspond to active cluster-heads and edges represent significant inter-cluster interaction or interference. This graph-based abstraction retains local autonomy while preserving the dominant couplings that govern relaying burden, coordination backlog, and interference propagation.

The clustered formulation also provides a clearer systems interpretation than either a fully centralized controller or a flat per-UAV controller. A centralized baseline remains useful because it quantifies the value of global information under the same cluster-level control semantics. However, deployment-facing execution still requires local decisions to be made without a hidden network-wide scheduling signal. Accordingly, the present work treats decentralized execution as a structural property of the controller rather than as a post hoc implementation detail.

D. Why Burst-Split Control Matters

Within each cluster burst, the controller must jointly regulate two distinct communication phases: intra-cluster service between the cluster-head and its subordinate UAVs, and inter-cluster coordination between neighboring cluster-heads. To capture this tradeoff explicitly, each cluster-head selects a local action

$$a_k(t) = (m_k(t), \rho_k(t)), \quad (3)$$

where $m_k(t)$ denotes the local MAC or hybrid-access decision and $\rho_k(t)$ controls how the burst duration t_{burst} is partitioned:

$$T_{1,k}(t) = \rho_k(t) t_{\text{burst}}, \quad (4)$$

$$T_{2,k}(t) = (1 - \rho_k(t)) t_{\text{burst}}. \quad (5)$$

Here, $T_{1,k}(t)$ is allocated to intra-cluster head-subordinate service, whereas $T_{2,k}(t)$ is allocated to inter-cluster head-head coordination. This burst split is central to the proposed formulation, because fixed partitioning is generally inadequate

when cluster size, backlog, graph degree, interference pressure, or failure-recovery demand vary over time. The controller must therefore learn not only *what* local access behavior to adopt, but also *how much* burst time to devote to service versus coordination.

E. Problem Perspective and Mathematical Scope

The resulting communication problem is best viewed as a decentralized graph-coupled control process. At burst time t , the global network state may be interpreted as

$$s_t = (X_t, V_t, Q_t, C_t, L_t, G_t, B_t^{\text{coord}}, B_t^{\text{ctrl}}, H_t), \quad (6)$$

where X_t and V_t denote UAV positions and velocities, Q_t denotes queue states, $C_t = \{C_k(t)\}$ is the cluster partition, $L_t = \{L_k(t)\}$ denotes the active cluster-heads, G_t is the cluster interference graph, B_t^{coord} and B_t^{ctrl} denote coordination and control backlogs, and H_t denotes leader-health states. The control law must operate under decentralized execution, meaning that each cluster-head acts only on its local observation and neighborhood summaries, even though the resulting transition dynamics remain coupled through mobility, interference, reassociation, split/merge events, and local leader handover.

This perspective is important because it moves the work away from an overly narrow MAC-selection view and toward a broader formulation of decentralized communication management in clustered FANETs. In this formulation, clustering, burst allocation, graph-aware coordination, and fault tolerance are not independent modules; they are coupled components of one control architecture.

F. Main Contributions

The contributions of this paper are summarized as follows.

- We formulate dynamic clustered FANET communication control as a decentralized graph-coupled decision problem in which cluster-heads act as the local agents and the network evolves through reassociation, split, merge, and local leader handover.
- We introduce an explicit burst-allocation variable that partitions each cluster burst between intra-cluster service and inter-cluster coordination, thereby exposing a physically meaningful tradeoff that is not captured by a global MAC-selection abstraction.
- We develop a graph-aware multi-agent reinforcement learning (MARL) controller with recurrent memory and dueling action evaluation for decentralized cluster-head control, while retaining a structured centralized MCA-D3QN single-agent reinforcement learning (SARL) baseline under the same cluster-level action interpretation.
- We incorporate local leader handover as a robustness extension and evaluate the architecture under topology variation, imperfect graph information, leader failures, and train-test mismatch conditions that are directly relevant to complexity-aware evaluation of aerial networking systems.

- We design the empirical study to cover not only offered-load benchmarking, but also burst-split behavior, topology sensitivity, cluster-dynamics sanity, leader-failure recovery, generalization across regimes, architectural ablations, and complexity-aware evaluation under simulated conditions.

The paper deliberately adopts conservative claims. It does not argue for universal optimality, nor does it present fault tolerance as the primary novelty. Instead, it focuses on a mathematically explicit decentralized formulation and on identifying the conditions under which graph-aware burst-split control is warranted in dynamic clustered FANETs.

G. Paper Organization

The remainder of the paper is organized as follows. Section II positions the paper with respect to decentralized wireless scheduling, graph-based wireless control, UAV/FANET reinforcement learning, and clustered aerial networking. Table I summarizes the principal notation used throughout the manuscript. Section III presents the network model, dynamic clustering process, graph construction, local observations and actions, reward structure, and decentralized problem formulation. Section IV introduces the graph-aware learning architecture, burst-action encoding, centralized baseline, and complexity interpretation. Section V defines the experimental setup, compared methods, metrics, and robustness protocols. Section VI reports the main empirical findings, including topology sensitivity, burst-split behavior, mismatch robustness, failure recovery, ablations, and deployability. Section VII discusses the implications and limitations of the present formulation, and Section VIII concludes the paper.

II. RELATED WORK

A. Decentralized Wireless Scheduling and Resource Allocation

Decentralized wireless scheduling has long been motivated by the mismatch between rapidly varying local channel conditions and the latency, signaling overhead, and scalability limits of centralized coordination [1]–[4]. Classical distributed MAC mechanisms provide lightweight contention resolution and practical medium access under partial information, but their control granularity is often limited and their decision structure is not explicitly designed to capture graph-coupled coordination among dynamically interacting subnetworks [5]–[7]. More recently, learning-based approaches have improved adaptability under partial observability across spectrum allocation, power control, and scheduling [4], [8], [9]. The centralized-training decentralized-execution (CTDE) paradigm has emerged as a practical framework for enabling coordination during training while preserving local-only execution [9], [10]. Nevertheless, many such methods still operate either at a flat node level or through a system abstraction that suppresses locality by reducing network behavior to a single global decision variable.

For the present problem, the main limitation of this literature is therefore not the absence of decentralization itself, but the

absence of a control abstraction that can jointly represent local service pressure and cross-group coordination demand within the same decision interval. In a clustered FANET, these two pressures coexist and evolve asynchronously across space and time. A formulation that ignores this structure risks reducing a graph-coupled communication-control problem to a coarse network-wide selector, thereby weakening both the physical interpretation of the action space and the meaning of decentralized execution.

B. Graph Learning for Wireless Control

Graph-based learning has become increasingly important in wireless control because interference, neighborhood coupling, and coordination dependencies are inherently relational and sparse [11]–[13]. Graph neural networks (GNNs) provide a natural mechanism for encoding local interaction structure without resorting to unstructured pooling of all observations. Eisen and Ribeiro [14] demonstrated optimal resource allocation using random-edge GNNs, while Shen et al. [13] established theoretical foundations for GNN-based scalable radio resource management. More recent work has shown that GNNs can unfold classical optimization algorithms for power control [15], assist massive MIMO power allocation [16], learn decentralized allocations across heterogeneous link topologies [17], and support distributed scheduling in multi-cell networks through heterogeneous graph representations [18]. Graph attention mechanisms are particularly attractive when neighbor influence is heterogeneous, enabling the controller to weight local interactions according to instantaneous relevance [12], [19]. Lee et al. [20] applied GNN-based scheduling specifically to multi-UAV D2D networks, demonstrating the relevance of graph representations in aerial communication settings.

However, graph learning alone does not determine the correct control abstraction. Prior work has used graph models at different levels of granularity, including individual links, radios, or generic node embeddings inside centralized pipelines [13], [14], [17]. The key modeling question is therefore not only how to encode graph structure, but also what the graph nodes, edges, and actions represent. In the present work, the graph is defined over dynamic cluster-heads, which are themselves the decentralized decision-making agents. This distinction is central. The graph is not merely a feature-extraction layer applied to a global controller; it is the structural object that defines inter-cluster coupling under decentralized execution.

C. MARL in UAV and FANET Communication Systems

Multi-agent reinforcement learning (MARL) has been extensively investigated for communication and control problems in wireless networks, particularly in settings characterized by partial observability, nonstationarity, and distributed decision making [4], [21]–[23]. Value decomposition methods such as VDN [24] and QMIX [25] provide factored value functions for cooperative settings, while MAPPO [26] has shown surprisingly strong performance in cooperative multi-agent games. CTDE-based DRL has been applied to multi-UAV

relay communication [10] and multi-agent spectrum access [8]. These studies establish the potential of learning-based decentralized adaptation, but they also highlight recurring difficulties, including unstable credit assignment, sensitivity to topology changes, and the gap between centralized training assumptions and decentralized deployment constraints [22], [23]. These challenges become more pronounced in aerial networks, where communication quality, relative geometry, and coordination demand can change rapidly with mobility.

Within UAV and FANET settings, prior work has explored learning-based trajectory design and resource management [27], cooperative multi-UAV networking [28], UAV swarm communication optimization [29], joint communication and computing in multi-UAV edge networks [30], DRL-based routing in FANETs [31], and cellular-connected UAV navigation [32]. Recent surveys provide comprehensive overviews of UAV communication architectures [33]–[35]. Nevertheless, a substantial portion of this literature still assumes either flat node sets, homogeneous contention conditions, or control abstractions that do not explicitly distinguish between local service and inter-group coordination. By contrast, the present work is organized around decentralized cluster-head control on a dynamic interference graph, with an explicit local burst-allocation variable that governs the tradeoff between intra-cluster communication and inter-cluster coordination. The distinction is not merely algorithmic; it changes the underlying decision semantics and the deployment interpretation of the controller.

D. Clustered and Hierarchical Aerial Networking

Clustering and hierarchical control have been studied extensively in mobile ad hoc and aerial networks as mechanisms for improving scalability, reducing control overhead, and localizing coordination [35]–[40]. More recently, Khan et al. [41] proposed adaptive clustering for delay-constrained FANET communication in IEEE TMC, while Arafat and Moh [42] surveyed cluster-based routing protocols for UAV networks. Yang et al. [43] provided a broader perspective on AI-empowered intelligent transportation systems that includes clustering approaches. Existing clustered approaches have primarily focused on cluster formation, leader election, routing support, topology maintenance, or mobility-aware hierarchy management [38]–[42]. These contributions are important because they demonstrate that hierarchy is often essential for scalable aerial networking. However, they do not by themselves provide a decentralized control formulation for burst-level communication management over a time-varying interference graph.

The present work builds on the clustered perspective while extending it in a more operational direction. First, cluster evolution is treated as part of the controlled environment through reassociation, split, merge, and local leader handover. Second, cluster-heads do not merely maintain hierarchy; they act as the decentralized agents that allocate communication effort between local service and cross-cluster coordination. In this sense, hierarchy is not an auxiliary preprocessing stage but a structural component of the control problem itself.

E. Fault Tolerance, Fairness, and Robust Decentralized Operation

A decentralized clustered architecture also raises two additional requirements that are often treated only peripherally in related work: robustness to local coordinator failure and fairness across dynamically evolving subnetworks. In hierarchical aerial systems, cluster-head or leader failure can create a local single point of disruption unless handover or re-election mechanisms are incorporated directly into the system design [44], [45]. Alsarhan et al. [44] specifically addressed fault-tolerant cluster-head election in FANETs, while Li et al. [45] proposed fault-tolerant communication schemes for multi-UAV cooperative networks. UAV-assisted emergency networks further motivate resilient architectures [46]. Likewise, decentralized adaptation can improve local efficiency while still producing uneven service across clusters unless fairness is explicitly monitored or regularized [47]. These issues are especially important in dynamic FANETs, where mobility, spatial imbalance, and coordination-critical traffic can magnify local instability.

The present work therefore treats local leader handover as a robustness extension of the decentralized control architecture and uses fairness-aware evaluation to assess whether localized adaptation is achieved without systematically starving disadvantaged clusters. This perspective differs from work in which robustness and fairness are treated only as secondary afterthoughts. Here, both are integrated into the problem interpretation and experimental design.

F. Gap Statement and Positioning of the Present Work

Taken together, the above literature shows strong progress in decentralized wireless scheduling [1], [2], [4], [9], graph-based wireless learning [13]–[15], [18], [19], MARL for dynamic communication systems [8], [10], [27]–[29], and clustered aerial networking [33], [34], [41], [42], [44]. Nevertheless, a clear gap remains at their intersection. Existing work does not yet provide a mathematically explicit decentralized clustered FANET formulation in which: i) the agent set is defined by dynamic cluster-heads rather than by a fixed global controller or a flat per-node abstraction; ii) inter-cluster coupling is represented through a time-varying cluster interference graph; and iii) each local action jointly governs access behavior and the allocation of burst time between intra-cluster service and inter-cluster coordination.

This paper addresses that gap. It does not claim that clustering, graph learning, MARL, fairness, or fault tolerance are individually new. Rather, it contributes a technically specific synthesis in which these elements are integrated into a single decentralized communication-control formulation. The resulting architecture combines dynamic clustering, graph-aware local decision making, adaptive burst allocation, structured centralized comparison, and robustness-oriented evaluation under mismatch, imperfect graph information, and local leader failures.

III. SYSTEM MODEL AND PROBLEM FORMULATION

Table I summarizes the principal symbols used in this section. The formulation is defined at the cluster-head level

TABLE I

KEY NOTATION USED THROUGHOUT THE DECENTRALIZED CLUSTERED FANET FORMULATION. THE SYMBOLS LISTED HERE ARE RESTRICTED TO THE CORE VARIABLES THAT APPEAR REPEATEDLY IN THE SYSTEM MODEL, PROPOSED METHOD, AND EXPERIMENTAL ANALYSIS.

Symbol	Meaning	Domain / Unit
n	Total number of UAVs in the FANET	Count
$\mathcal{U}(t)$	Set of active UAVs at decision instant t	Set
$c(t)$	Number of active clusters at decision instant t	Count
$C_k(t)$	k th cluster at decision instant t	Set
$L_k(t)$	Cluster-head (leader) of cluster $C_k(t)$	UAV index
$G_t = (V_t, E_t)$	Dynamic cluster interference graph, where vertices correspond to cluster-heads and edges represent significant inter-cluster interaction	Graph
$o_k(t)$	Local observation available to cluster-head k at decision instant t	Vector
$a_k(t)$	Local control action selected by cluster-head k	Tuple
$m_k(t)$	Local medium-access or hybrid-access decision of cluster-head k	Discrete mode
$\rho_k(t)$	Burst-allocation ratio governing the fraction of burst time assigned to intra-cluster service	$[0, 1]$
t_{burst}	Total duration of one cluster communication burst	Time
$T_{1,k}(t)$	Intra-cluster communication duration allocated to head-subordinate exchange in cluster k	Time
$T_{2,k}(t)$	Inter-cluster coordination duration allocated to head-head exchange for cluster k	Time
$r_k(t)$	Local reward associated with cluster-head k at decision instant t	Scalar
$R_{\text{team}}(t)$	Team-level objective obtained by aggregating local rewards and fairness regularization	Scalar
$J(t)$	Jain's fairness index computed over per-cluster or per-flow utility values	Scalar
R_c	Cluster-association radius or equivalent cluster-membership threshold parameter	Distance / policy parameter
R_I	Inter-cluster interference radius used in graph construction or interaction modeling	Distance
τ_I	Interference, adjacency, or interaction threshold used when forming inter-cluster graph edges	Threshold
$H_k(t)$	Health state or reliability indicator of cluster-head k used for local handover decisions	Scalar / state variable

because the objective is decentralized execution over a dynamic clustered FANET rather than flat per-UAV control or a network-wide broadcast scheduler.

A. Network Model

Consider a FANET composed of n UAVs. Let

$$\mathcal{U}(t) = \{1, 2, \dots, n\} \quad (7)$$

denote the UAV index set at decision instant t . The UAV cardinality is fixed in the nominal setting, but the communica-

tion structure evolves because of mobility, queue fluctuations, interference variation, and cluster reconfiguration.

Let the latent network state at decision instant t be denoted by

$$s_t = (X_t, V_t, Q_t, C_t, L_t, G_t, B_t^{\text{coord}}, B_t^{\text{ctrl}}, H_t), \quad (8)$$

where:

- X_t and V_t are the UAV position and velocity states,
- Q_t is the queue-state collection,
- $C_t = \{C_k(t)\}_{k=1}^{c(t)}$ is the cluster partition,
- $L_t = \{L_k(t)\}_{k=1}^{c(t)}$ is the set of active cluster-heads,
- $G_t = (V_t, E_t)$ is the dynamic cluster interference graph,
- B_t^{coord} and B_t^{ctrl} denote coordination and control backlogs,
- H_t denotes leader-health states.

The controller is not assumed to observe s_t directly.

B. Dynamic Cluster Partition

At each decision instant t , the UAV set is partitioned into $c(t)$ active clusters:

$$\mathcal{U}(t) = \bigcup_{k=1}^{c(t)} C_k(t). \quad (9)$$

The partition must satisfy both disjointness and unique membership:

$$C_k(t) \cap C_\ell(t) = \emptyset, \quad k \neq \ell, \quad (10)$$

$$\sum_{k=1}^{c(t)} \mathbf{1}\{i \in C_k(t)\} = 1, \quad \forall i \in \mathcal{U}(t). \quad (11)$$

Each cluster has one cluster-head

$$L_k(t) \in C_k(t), \quad k = 1, \dots, c(t), \quad (12)$$

and the active cluster-heads define the decentralized agent set.

This representation determines the control granularity. Subordinate UAVs do not select communication actions directly; instead, their state contributes to the local observation of the corresponding cluster-head.

C. Cluster Membership and Reconfiguration

Cluster membership is determined by a score that captures proximity, communication quality, mobility compatibility, and cluster load. For UAV i and candidate cluster k , define

$$S_{i,k}(t) = w_d f_d(i, k, t) + w_s f_s(i, k, t) + w_m f_m(i, k, t) + w_q f_q(i, k, t), \quad (13)$$

where f_d , f_s , f_m , and f_q are normalized feature terms and $w_d, w_s, w_m, w_q \geq 0$ are the corresponding weights. The nominal assignment is

$$\chi_i(t) = \arg \max_{1 \leq k \leq c(t)} S_{i,k}(t), \quad (14)$$

subject to the cluster-association radius or threshold parameter R_c and a hysteresis rule.

Let $u = \chi_i(t^-)$ denote the current cluster of UAV i . Reassociation from cluster u to cluster v occurs only if

$$S_{i,v}(t) - S_{i,u}(t) \geq \Delta_{\text{reasoc}}, \quad (15)$$

where $\Delta_{\text{reasoc}} > 0$ is a design margin introduced to suppress oscillatory reassignment.

Cluster evolution further includes split and merge events. These are modeled through thresholded predicates:

$$\begin{aligned} \text{split}(C_k(t)) \text{ if } |C_k(t)| > N_{\text{split}}^{\max} \\ \text{and } \Psi_k^{\text{split}}(t) = 1, \end{aligned} \quad (16)$$

$$\begin{aligned} \text{merge}(C_k(t), C_\ell(t)) \text{ if } |C_k(t)| < N_{\text{merge}}^{\min} \\ \text{and } \Psi_{k\ell}^{\text{merge}}(t) = 1. \end{aligned} \quad (17)$$

where $\Psi_k^{\text{split}}(t)$ and $\Psi_{k\ell}^{\text{merge}}(t)$ are admissibility predicates that may depend on geometry, stability, or compatibility. This form captures the implemented threshold-and-predicate logic without prematurely fixing scenario-specific constants.

D. Dynamic Cluster Interference Graph

The interdependence among cluster-heads is represented by a dynamic cluster interference graph

$$G_t = (V_t, E_t), \quad (18)$$

with node set

$$V_t = \{1, 2, \dots, c(t)\}, \quad (19)$$

where node k corresponds to cluster-head $L_k(t)$. An undirected edge $(k, \ell) \in E_t$ indicates that clusters k and ℓ are sufficiently coupled to affect one another through interference, coordination demand, or relay interaction. A generic edge condition is

$$(k, \ell) \in E_t \iff d_{k\ell}(t) \leq R_I \text{ or } \hat{\gamma}_{k\ell}(t) \leq \tau_I, \quad (20)$$

where $d_{k\ell}(t)$ is the inter-head distance, R_I is the inter-cluster interaction radius, and $\hat{\gamma}_{k\ell}(t)$ is an interference or link-quality proxy compared against threshold τ_I .

The graph is the structural object through which decentralized coupling enters the control problem: even under local execution, the action of one cluster-head may affect neighboring clusters through contention, coordination overlap, or relaying.

E. Burst-Time Structure and Local Control Variables

Each decision step corresponds to one communication burst of duration t_{burst} . Cluster-head k selects a local action

$$a_k(t) = (m_k(t), \rho_k(t)), \quad (21)$$

where $m_k(t)$ denotes the local access mode and $\rho_k(t) \in [0, 1]$ is the burst-allocation ratio. The burst budget is partitioned as

$$t_{\text{burst}} = T_{1,k}(t) + T_{2,k}(t), \quad (22)$$

with

$$T_{1,k}(t) = \rho_k(t) t_{\text{burst}}, \quad (23)$$

$$T_{2,k}(t) = (1 - \rho_k(t)) t_{\text{burst}}. \quad (24)$$

For continuity, the burst-allocation ratio is equivalently written as

$$\rho_k(t) = \frac{T_{1,k}(t)}{t_{\text{burst}}}, \quad (25)$$

which makes explicit that $\rho_k(t)$ is the normalized fraction of burst time assigned to the intra-cluster phase.

Here, $T_{1,k}(t)$ is the head-subordinate communication phase for cluster k , while $T_{2,k}(t)$ is the head-head coordination phase used for schedule exchange, relay signaling, topology updates, and handover-related control. The decentralized network action at time t is therefore

$$\mathbf{a}(t) = \{a_k(t)\}_{k=1}^{c(t)}. \quad (26)$$

No network-wide broadcast action replaces (26); decentralized execution is preserved by construction.

F. Local Observation Model

Cluster-head k acts on a local observation vector

$$\mathbf{o}_k(t) = [\phi_k^{\text{loc}}(t), \phi_k^{\text{graph}}(t), \phi_k^{\text{queue}}(t), \phi_k^{\text{health}}(t), \phi_k^{\text{hist}}(t)], \quad (27)$$

where:

- $\phi_k^{\text{loc}}(t)$ summarizes local cluster service state,
- $\phi_k^{\text{graph}}(t)$ summarizes neighboring-cluster interaction state,
- $\phi_k^{\text{queue}}(t)$ encodes backlog and coordination demand,
- $\phi_k^{\text{health}}(t)$ encodes leader-health and handover indicators,
- $\phi_k^{\text{hist}}(t)$ encodes recent burst-allocation history.

In practice, $\mathbf{o}_k(t)$ may include cluster size, queue intensity, local delay and collision proxies, graph degree, neighbor backlog summaries, leader stability indicators, and the previously used burst split. The exact feature inventory can be specialized later without changing the role of the observation model.

Let $P(\cdot)$ denote the environment transition kernel. The system evolves according to

$$s_{t+1} \sim P(s_{t+1} | s_t, \mathbf{a}(t)), \quad (28)$$

where the transition is induced jointly by mobility, queue evolution, clustering dynamics, graph evolution, and burst-level execution. No closed-form transition model is assumed.

G. Reward and Team Objective

The decentralized controller is trained using local cluster-level rewards. Let $\tilde{T}_k^{\text{intra}}(t)$ and $\tilde{T}_k^{\text{inter}}(t)$ denote normalized intra-cluster and inter-cluster throughput terms, let $\tilde{D}_k^{\text{intra}}(t)$ and $\tilde{D}_k^{\text{inter}}(t)$ denote normalized delay terms, and let $\tilde{C}_k(t)$, $\tilde{I}_k(t)$, $\tilde{Q}_k(t)$, $\tilde{E}_k(t)$, and $\tilde{F}_k(t)$ denote normalized penalties for collisions, interference, queue stress, energy expenditure, and robustness-related disruption. The local reward is

$$\begin{aligned} r_k(t) = & \alpha_1 \tilde{T}_k^{\text{intra}}(t) + \alpha_2 \tilde{T}_k^{\text{inter}}(t) \\ & - \beta_1 \tilde{D}_k^{\text{intra}}(t) - \beta_2 \tilde{D}_k^{\text{inter}}(t) \\ & - \eta_1 \tilde{C}_k(t) - \eta_2 \tilde{I}_k(t) \\ & - \eta_3 \tilde{Q}_k(t) - \eta_4 \tilde{E}_k(t) - \eta_5 \tilde{F}_k(t). \end{aligned} \quad (29)$$

where all coefficients are nonnegative.

To support team-level analysis and optional regularization, define

$$R_{\text{team}}(t) = \frac{1}{c(t)} \sum_{k=1}^{c(t)} r_k(t) + \lambda_J J(t), \quad (30)$$

where $\lambda_J \geq 0$ weights the fairness term and

$$J(t) = \frac{\left(\sum_{k=1}^{c(t)} x_k(t) \right)^2}{c(t) \sum_{k=1}^{c(t)} x_k^2(t)} \quad (31)$$

is Jain's fairness index computed from a nonnegative per-cluster serviced-throughput quantity $x_k(t)$. In the present study, $J(t)$ is used both as an evaluation metric and, when enabled, as a team-regularization term.

H. Fault-Tolerant Cluster-Head Handover

Fault tolerance is modeled as a local robustness extension. Let $\Omega_k^{\text{fail}}(t) \in \{0, 1\}$ indicate a leader-failure or handover-trigger event for cluster k . Candidate replacements are ranked by a health score

$$H_u(t) = \omega_1 \tilde{e}_u(t) + \omega_2 \tilde{d}_u(t) + \omega_3 \tilde{s}_u(t) - \omega_4 \tilde{q}_u(t), \quad (32)$$

where $\tilde{e}_u(t)$ is a normalized residual-energy term, $\tilde{d}_u(t)$ is a connectivity-related term, $\tilde{s}_u(t)$ is a mobility-stability term, and $\tilde{q}_u(t)$ reflects service burden. When handover is triggered, the next leader is selected locally as

$$L_k(t^+) = \arg \max_{u \in C_k(t) \setminus \{L_k(t)\}} H_u(t). \quad (33)$$

This preserves decentralized execution because leader replacement is determined within the affected cluster rather than by a network-wide control action.

I. Dec-POMDP and CTDE Interpretation

The overall problem can be interpreted as a decentralized partially observable Markov decision process (Dec-POMDP),

$$\mathcal{M} = \langle \mathcal{S}, \{\mathcal{A}_k\}_{k=1}^{c_{\max}}, \{\mathcal{O}_k\}_{k=1}^{c_{\max}}, P, \{r_k\}_{k=1}^{c_{\max}}, \Omega, \gamma \rangle, \quad (34)$$

where $c_{\max}$ is an upper bound used to accommodate variable cluster count through padding and masking, $\mathcal{S}$ is the latent state space, $\mathcal{A}_k$ is the local action space, $\mathcal{O}_k$ is the local observation space, P is the transition kernel, Ω is the observation model, and $\gamma \in (0, 1)$ is the discount factor.

Under decentralized execution, cluster-head k selects its action from its local observation history

$$\tau_k(t) = \{\mathbf{o}_k(0), \dots, \mathbf{o}_k(t)\}, \quad (35)$$

according to

$$a_k(t) \sim \pi_{\theta,k}(\cdot \mid \tau_k(t)). \quad (36)$$

The same setting also supports centralized-training/decentralized-execution (CTDE) baselines, provided that any additional joint information is used only during training and not at execution time. This distinction is maintained explicitly so that decentralized execution remains structurally real rather than nominal.

IV. PROPOSED METHOD

This section develops the controller that operates on the decentralized clustered FANET model introduced in Section III. The objective is to construct a policy whose state representation, action space, learning rule, and execution mechanism remain consistent with cluster-head-level decentralized control over a dynamic interference graph.

A. Overall Decentralized Architecture

At each burst decision instant t , the controller performs four coupled operations:

- 1) update the cluster structure through reassociation, split, merge, and leader-status changes;
- 2) construct the dynamic cluster interference graph $G_t = (V_t, E_t)$ from the active cluster-head set;
- 3) form local cluster-head observations $\mathbf{o}_k(t)$ from cluster state, neighbor summaries, and recent execution history;
- 4) select and execute the decentralized local actions

$$a_k(t) = (m_k(t), \rho_k(t)), \quad k = 1, \dots, c(t).$$

The execution rule is decentralized: once the graph and local observations are formed, no network-wide action aggregator replaces the local decisions. Coupling is handled only through the graph structure and the coordination phase embedded in the burst.

B. Dynamic Clustering and Graph Construction

Let

$$A_{k\ell}(t) = \begin{cases} 1, & (k, \ell) \in E_t, \\ 0, & \text{otherwise,} \end{cases} \quad (37)$$

denote the adjacency matrix of the cluster-head graph, and let

$$\mathcal{N}_k(t) = \{\ell \in V_t : A_{k\ell}(t) = 1\} \quad (38)$$

be the neighbor set of cluster-head k . The graph is updated after each clustering event so that reassociation, split, merge, and local handover are reflected immediately in the active node set and edge structure.

This modeling choice is deliberate. The graph is defined over cluster-heads rather than all UAVs, which yields two benefits:

- it preserves the deployment-facing interpretation that cluster-heads are the actual decentralized decision makers;
- it controls action-space growth while retaining the dominant inter-cluster couplings relevant to interference, coordination, and relaying.

Thus, the graph is not an auxiliary embedding device layered on top of a centralized action space; it is the structural object that defines decentralized interaction.

C. Graph-Aware MARL Policy

The proposed controller is a multi-agent graph-attention recurrent dueling Q-network, denoted MAGAT-D3QN. It combines graph message passing, temporal memory, and dueling value decomposition within a cluster-head policy. The purpose of the graph module is to encode heterogeneous inter-cluster coupling; the purpose of the recurrent module is to capture short-horizon temporal dependence under partial observability; and the purpose of the dueling Q-structure is to stabilize local action ranking when multiple actions produce similar returns.

Let $\mathbf{h}_k^{(0)}(t)$ denote the initial latent embedding obtained from $\mathbf{o}_k(t)$.

$$e_{kq}^{(\ell)}(t) = \text{LReLU}\left(\mathbf{a}^{(\ell)\top} \left[W^{(\ell)} \mathbf{h}_k^{(\ell)}(t) \parallel W^{(\ell)} \mathbf{h}_q^{(\ell)}(t) \right]\right) \quad (39)$$

$$\alpha_{kq}^{(\ell)}(t) = \frac{\exp\left(e_{kq}^{(\ell)}(t)\right)}{\sum_{j \in \mathcal{N}_k(t) \cup \{k\}} \exp\left(e_{kj}^{(\ell)}(t)\right)} \quad (40)$$

where $\text{LReLU}(\cdot)$ denotes the LeakyReLU activation and $\parallel$ denotes concatenation. The graph-updated embedding for cluster-head k is then obtained by multi-head aggregation:

$$\mathbf{h}_k^{(\ell+1)}(t) = \sigma \left(\frac{1}{H_{\text{att}}} \sum_{p=1}^{H_{\text{att}}} \sum_{q \in \mathcal{N}_k(t) \cup \{k\}} \alpha_{kq}^{(\ell,p)}(t) W^{(\ell,p)} \mathbf{h}_q^{(\ell)}(t) \right), \quad (41)$$

where H_{att} is the number of attention heads and $\sigma(\cdot)$ is a nonlinear activation.

Equation (41) implies that the embedding of cluster-head k is updated by a weighted aggregation of its own state and the states of its neighbors. The coefficients in (40) adapt these weights to the current graph context. Hence, the graph module does not impose uniform neighbor influence; it learns which neighboring clusters matter more at the current decision instant. Multi-head attention is used only as a representation mechanism. The attention heads are not separate actors and do not alter the decentralized semantics of the action space.

Because the burst-control problem is temporally correlated, the post-message-passing embedding is processed by a recurrent memory block:

$$\mathbf{z}_k(t) = \text{GRU}(\mathbf{h}_k(t), \mathbf{z}_k(t-1)), \quad (42)$$

where $\mathbf{h}_k(t)$ denotes the graph-updated embedding and $\mathbf{z}_k(t)$ is the hidden memory state. The GRU preserves short-horizon temporal context, including queue buildup, recent burst-allocation history, and transients induced by handover or rapid topology change.

The final local action-value estimate is obtained through a dueling decomposition:

$$\bar{A}_k(\mathbf{z}_k(t)) = \frac{1}{|\mathcal{A}_k|} \sum_{u' \in \mathcal{A}_k} A_k(\mathbf{z}_k(t), u') \quad (43)$$

$$Q_k(\mathbf{o}_k(t), u) = V_k(\mathbf{z}_k(t)) + A_k(\mathbf{z}_k(t), u) - \bar{A}_k(\mathbf{z}_k(t)) \quad (44)$$

where $V_k(\cdot)$ is the state-value stream, $A_k(\cdot, \cdot)$ is the advantage stream, and u is a discrete action index. This decomposition separates the value of the local state from the relative utility of candidate actions under that state.

D. Burst-Allocation Action Encoding

The local action space is the Cartesian product

$$\mathcal{A}_k = \mathcal{M}_k \times \mathcal{P}_k, \quad (45)$$

where $\mathcal{M}_k$ is the discrete MAC-mode set and $\mathcal{P}_k$ is the feasible burst-ratio set. For implementation, each pair $(m_k(t), \rho_k(t))$ is encoded into a discrete index

$$u_k(t) = \text{enc}(m_k(t), \rho_k(t)), \quad (46)$$

with inverse mapping

$$(m_k(t), \rho_k(t)) = \text{dec}(u_k(t)). \quad (47)$$

This encoding allows a value-based MARL controller to optimize the joint local action without introducing an artificial factorization between MAC choice and burst allocation. The feasible ratio set $\mathcal{P}_k$ is constrained so that both communication phases remain nonzero:

$$T_{1,k}(t) \geq T_{1,\min}, \quad T_{2,k}(t) \geq T_{2,\min}, \quad (48)$$

which prevents pathological allocations that eliminate either local service or inter-cluster coordination.

E. Centralized MCA-D3QN Baseline

The centralized MCA-D3QN baseline is used as a structured centralized SARL comparator. It observes a padded joint cluster-level state

$$\mathbf{o}^{\text{cen}}(t) = [\mathbf{o}_1^\top(t), \mathbf{o}_2^\top(t), \dots, \mathbf{o}_{c_{\max}}^\top(t)]^\top, \quad (49)$$

where inactive cluster slots are masked, and outputs the joint cluster-level action vector

$$\mathbf{a}^{\text{cen}}(t) = [a_1(t), a_2(t), \dots, a_{c_{\max}}(t)]. \quad (50)$$

Its role is methodological rather than architectural: it quantifies the benefit of global state access under the same cluster-level control semantics. It is therefore not a pooled broadcast wrapper and should not be interpreted as a global MAC selector.

F. Training Objective and Execution Rule

For the graph-aware controller, the primary learning signal is the local reward $r_k(t)$ defined in Section III-G. A representative temporal-difference loss for cluster-head k is

$$\mathcal{L}_{Q,k}(\theta) = \mathbb{E} \left[(y_k(t) - Q_k(\mathbf{o}_k(t), u_k(t); \theta))^2 \right], \quad (51)$$

with target

$$y_k(t) = r_k(t) + \gamma Q_k^-(\mathbf{o}_k(t+1), u_k^*(t+1); \bar{\theta}), \quad (52)$$

where

$$u_k^*(t+1) = \arg \max_{u' \in \mathcal{A}_k} Q_k(\mathbf{o}_k(t+1), u'; \theta) \quad (53)$$

is the online-network action used in the Double-DQN target, and $\bar{\theta}$ denotes the target-network parameters.

When team regularization is enabled, the overall optimization objective becomes

$$\mathcal{L}_{\text{tot}}(\theta) = \sum_{k=1}^{c(t)} \mathcal{L}_{Q,k}(\theta) + \lambda_{\text{reg}} \mathcal{L}_{\text{team}}(\theta), \quad (54)$$

where $\mathcal{L}_{\text{team}}(\theta)$ is derived from the team objective in (30). This term is used only as a regularizer; it does not replace the local reward structure.

At execution time, cluster-head k chooses

$$\hat{u}_k(t) = \arg \max_{u \in \mathcal{A}_k} Q_k(\mathbf{o}_k(t), u; \theta), \quad (55)$$

and the physical control action is recovered through

$$\hat{a}_k(t) = \text{dec}(\hat{u}_k(t)). \quad (56)$$

Thus, the learned policy remains local at execution time: once observations and graph context are formed, no centralized executor is required.

G. Complexity and Communication Overhead

The proposed controller is intended to be deployment-facing, so both analytic and measured complexity are relevant. Let d_h denote the latent width and let H_{att} denote the number of graph-attention heads. A representative graph-message-passing cost per decision step is

$$\mathcal{C}_{\text{GAT}}(t) = \mathcal{O}(H_{\text{att}} |E_t| d_h + c(t) d_h^2), \quad (57)$$

while the recurrent and action-head costs are

$$\mathcal{C}_{\text{GRU}}(t) = \mathcal{O}(c(t) d_h^2), \quad (58)$$

$$\mathcal{C}_{\text{head}}(t) = \mathcal{O}(c(t) d_h |\mathcal{A}_k|). \quad (59)$$

Communication overhead is induced by neighbor-summary exchange during the coordination phase. If b_{sum} denotes the directed summary payload in bytes transmitted from one cluster-head to one neighbor per burst, then the system-level communication overhead is

$$B_{\text{comm}}(t) = b_{\text{sum}} \sum_{k=1}^{c(t)} |\mathcal{N}_k(t)| = b_{\text{sum}} c(t) \bar{d}(t), \quad (60)$$

where $\bar{d}(t)$ is the average cluster-graph degree.

Equations (57)–(60) provide a first-order analytic view of cost scaling with graph density, latent width, and action-space size. In the empirical study, this analysis is complemented by measured parameter count, model memory, average and tail inference latency, decision overhead per active cluster, and realized communication overhead.

ALGORITHM PLACEHOLDERS

V. EXPERIMENTAL SETUP

This section defines the simulation environment, evaluation metrics, hyperparameter tuning protocol, and failure-injection model used throughout the empirical study to the level of detail required for independent reproduction. All configuration parameters are summarized in Table II, and the training and evaluation protocol is specified in Table IV. Table V lists the compared methods.

Algorithm 1 Dynamic Clustering and Local Leader Update

Require: UAV set $\mathcal{U}(t)$, previous cluster partition $\{C_k(t-1)\}$, previous leaders $\{L_k(t-1)\}$, membership-score model $S_{i,k}(t)$

Ensure: Updated clusters $\{C_k(t)\}$, updated leaders $\{L_k(t)\}$, updated graph $G_t = (V_t, E_t)$

- 1: **for** each UAV $i \in \mathcal{U}(t)$ **do**
- 2: Evaluate admissible membership scores $\{S_{i,k}(t)\}$.
- 3: Update provisional assignment $\chi_i(t)$ using the reassociation rule in (15).
- 4: **end for**
- 5: Form provisional clusters $\{C_k(t)\}$ from $\{\chi_i(t)\}$.
- 6: **for** each active cluster $C_k(t)$ **do**
- 7: **if** $|C_k(t)| > N_{\text{split}}^{\text{max}}$ and $\Psi_k^{\text{split}}(t) = 1$ **then**
- 8: Split $C_k(t)$.
- 9: **end if**
- 10: **if** $|C_k(t)| < N_{\text{merge}}^{\text{min}}$ and $\Psi_{k\ell}^{\text{merge}}(t) = 1$ for some ℓ **then**
- 11: Merge $C_k(t)$ with a compatible cluster.
- 12: **end if**
- 13: **if** $\Omega_k^{\text{fail}}(t) = 1$ **then**
- 14: Update $L_k(t)$ using (33).
- 15: **end if**
- 16: **end for**
- 17: Construct the cluster interference graph $G_t = (V_t, E_t)$ using (20).
- 18: **return** $\{C_k(t)\}$, $\{L_k(t)\}$, and G_t .

Algorithm 2 Decentralized Burst-Allocation Decision and Execution

Require: Active cluster-head set V_t , graph $G_t = (V_t, E_t)$, local observations $\{\mathbf{o}_k(t)\}_{k \in V_t}$

Ensure: Executed actions $\{a_k(t)\}_{k \in V_t}$, updated burst statistics, local rewards $\{r_k(t)\}_{k \in V_t}$

- 1: **for** each active cluster-head $k \in V_t$ **do**
- 2: Compute the local action index

$$\hat{u}_k(t) = \arg \max_{u \in \mathcal{A}_k} Q_k(\mathbf{o}_k(t), u).$$
- 3: Decode

$$\hat{a}_k(t) = \text{dec}(\hat{u}_k(t)) = (m_k(t), \rho_k(t)).$$
- 4: Compute burst durations

$$T_{1,k}(t) = \rho_k(t) t_{\text{burst}}, \quad T_{2,k}(t) = (1 - \rho_k(t)) t_{\text{burst}}.$$
- 5: **end for**
- 6: Execute intra-cluster service over $\{T_{1,k}(t)\}_{k \in V_t}$.
- 7: Execute inter-cluster coordination over $\{T_{2,k}(t)\}_{k \in V_t}$ subject to graph-coupled interactions on G_t .
- 8: Update local queues, coordination backlogs, graph-dependent statistics, and rewards $\{r_k(t)\}_{k \in V_t}$.
- 9: **return** $\{a_k(t)\}_{k \in V_t}$, updated environment state, and $\{r_k(t)\}_{k \in V_t}$.

Algorithm 3 Training and Evaluation Workflow

Require: Training seed set $\mathcal{S}_{\text{tr}}$, evaluation seed set $\mathcal{S}_{\text{ev}}$, scenario set $\mathcal{E}$, compared method set $\mathcal{M}$

Ensure: Aggregated manuscript-ready metrics

- 1: **for** each method $\mu \in \mathcal{M}$ **do**
- 2: Tune hyperparameters on training data only, if applicable.
- 3: Train the controller of method μ and save the selected checkpoint.
- 4: **for** each scenario $e \in \mathcal{E}$ **do**
- 5: **for** each seed $s \in \mathcal{S}_{\text{ev}}$ **do**
- 6: Execute the trained controller under scenario e and seed s .
- 7: Log network metrics, cluster-dynamics metrics, robustness metrics, and complexity metrics.
- 8: **end for**
- 9: **end for**
- 10: Aggregate results over seeds as mean $\pm$ standard deviation.
- 11: **end for**
- 12: Export the study-specific tables and figures for the final results section.

A. Simulation Environment

The simulator models a three-dimensional FANET operating over a $200 \times 200 \times 50$ m bounding volume with $n = 50$ UAVs. Each simulation episode spans $T_{\text{ep}} = 100$ decision steps, where each step corresponds to one communication burst of duration $t_{\text{burst}} = 0.5$ s (total episode time: 50 s). The simulation time resolution is $\Delta t_{\text{slot}} = 9 \mu\text{s}$.

PHY and Channel Model: The physical layer operates at a nominal data rate of $R = 5$ Mbps with QPSK modulation. Transmit power is $P_{\text{tx}} = 20$ dBm and the noise floor is $N_0 = -80$ dBm. The maximum acceptable bit-error rate is $\text{BER}_{\text{max}} = 10^{-3}$. The communication range is $R_{\text{comm}} = 200$ m. Path loss follows a free-space model with exponent $\eta = 2.0$ and constant $K_{\text{PL}} = 10^{-5}$:

$$\text{PL}(d) = K_{\text{PL}} \cdot d^\eta. \quad (61)$$

The default channel model is Nakagami- m fading with shape $m = 2.0$ and spread $\Omega = 1.0$. Fading ablations under AWGN (deterministic), Rician ($K_{\text{Ric}} = 3.0$ dB), and Rayleigh ($m = 1$) channels are reported separately. Propagation delay is computed as $\tau_{\text{prop}} = d/c$.

Packet Model and Traffic Generation: Each packet has a fixed payload of $L_{\text{pkt}} = 1000$ bytes. Under the default *smooth* traffic profile, packet arrivals follow a Poisson process at rate λ packets per second (pps) per UAV with memoryless exponential inter-arrival times. The offered-load sweep varies λ across 20 equally spaced steps from 50 pps to 1,000 pps. The default evaluation load is $\lambda = 525$ pps.

Two stress-test profiles are also evaluated:

- **Bursty on-off:** With probability $p_{\text{on}} = 0.30$, the per-step offered load is multiplied by a burst factor of $2.0 \times$; otherwise it is scaled to $0.35 \times$.

- **Heavy-tail:** The per-step load is drawn as $\lambda_{\text{eff}} = \lambda \cdot (\text{Pareto}(\alpha=1.8) + 1) \cdot s$, where $s = 1.0$ is a scale parameter, clipped to $[0.2\lambda, 4.0\lambda]$.

Queue and Drop Model: Each UAV maintains a FIFO queue with capacity $Q_{\text{max}} = 100$ packets. Packets arriving when the queue is full are dropped (drop-tail). Per-step queue evolution for member i in cluster k is

$$q_i(t+1) = \text{clip}(q_i(t) + a_i(t) - s_i(t) + \delta_i^{\text{ctrl}}(t), 0, Q_{\text{max}}), \quad (62)$$

where $a_i(t)$ is the number of arriving packets, $s_i(t)$ is the number of served packets, and $\delta_i^{\text{ctrl}}(t)$ is a control backpressure term from unserved local-control demand. The total drop count $\bar{\Pi}_{\text{drop}}$ reported in the results is the sum of queue-overflow drops and MAC-layer drops (failed transmissions after retry exhaustion) across all UAVs and steps.

CSMA/CA Sub-Simulator: When MAC mode $m_k(t) = 1$ (CSMA/CA), the intra-cluster sub-phase $T_{1,k}(t)$ is simulated with the following parameters: contention window $\text{CW}_{\text{min}} = 15$, $\text{CW}_{\text{max}} = 1023$; DIFS = 4 slots; SIFS = 2 slots; ACK timeout = 10 slots; maximum retry count = 1; RTS/CTS enabled (RTS = 3 slots, CTS = 3 slots, ACK = 2 slots). Collisions are counted as simultaneous-transmission events within the sub-phase.

TDMA Sub-Simulator: When MAC mode $m_k(t) = 0$ (TDMA), the intra-cluster sub-phase allocates equal-length slots to each member with a guard interval of $\Delta t_{\text{guard}} = 1 \mu\text{s}$ between slots.

Action Space: The local action $a_k(t) = (m_k(t), \rho_k(t))$ combines a discrete MAC-mode selection $m_k(t) \in \{0 : \text{TDMA}, 1 : \text{CSMA/CA}\}$ with a discretized burst-allocation ratio. The candidate ρ levels are

$$\mathcal{P}_{\text{cand}} = \{0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90\}. \quad (63)$$

A candidate level ρ is *valid* if both $T_{1,k} = \rho \cdot t_{\text{burst}} \geq T_{1,\text{min}}^{\text{min}}$ and $T_{2,k} = (1-\rho) \cdot t_{\text{burst}} \geq T_{2,\text{min}}^{\text{min}}$, where $T_{1,\text{min}}^{\text{min}} = T_{2,\text{min}}^{\text{min}} = 0.05$ s. Under the default $t_{\text{burst}} = 0.5$ s, all nine levels satisfy these constraints, yielding $|\mathcal{P}_k| = 9$ valid levels. The total action space per cluster-head is

$$|\mathcal{A}_k| = |\mathcal{M}_k| \times |\mathcal{P}_k| = 2 \times 9 = 18. \quad (64)$$

Action encoding uses $a_{\text{id}} = m_k \cdot |\mathcal{P}_k| + \rho_{\text{id}\text{x}}$ and decoding uses $\rho_{\text{id}\text{x}} = a_{\text{id}} \bmod |\mathcal{P}_k|$, $m_k = \lfloor a_{\text{id}} / |\mathcal{P}_k| \rfloor \bmod |\mathcal{M}_k|$.

Inter-Cluster Coordination Model: Coordination between neighboring cluster-heads uses a tail-aligned synchronization protocol with a coordination guard time of $\Delta t_{\text{coord}} = 0.05$ s. The temporal overlap between clusters k and j is

$$\text{overlap}(k, j) = \max(0, t_{\text{burst}} - \max(T_{1,k}, T_{1,j}) - \Delta t_{\text{coord}}). \quad (65)$$

Coordination capacity per edge is $C_{\text{coord}} = 4.0$ units/s, relay service capacity is $C_{\text{relay}} = 3.0$ units/s, and local control capacity is $C_{\text{local}} = 6.0$ units/s. The coordination success ratio between two neighbors is

$$\text{CSR}(k, j) = \text{clip}\left(\frac{\text{overlap}(k, j)}{\max(\min(T_{2,k}, T_{2,j}) - \Delta t_{\text{coord}}, 10^{-9})}, 0, 1\right). \quad (66)$$

Interference Model: When two neighboring cluster-heads (k, j) both select CSMA/CA ($m_k=m_j=1$), the interference penalty is $0.15 \cdot (1.25 - \text{CSR})$. When both select the same MAC mode (both TDMA), the penalty is $0.06 \cdot (1.25 - \text{CSR})$. For mixed MAC modes, the penalty is $0.02 \cdot (1.25 - \text{CSR})$. These penalties accumulate over all interfering neighbors.

Reward Normalization and Computation: Each reward component is independently normalized to $[0, 1]$ by clipping against fixed ceilings before weighted combination. The normalization ceilings are: $T_{\max} = 5.0$ Mbps, $D_{\max} = 200$ ms, $C_{\max} = 500$ collisions, $E_{\max} = 1.0$ energy units. The full local reward for cluster-head k at step t is

$$r_k(t) = \alpha_1 \hat{T}_k^{\text{local}} + \alpha_2 \hat{T}_k^{\text{inter}} - \beta_1 \hat{D}_k^{\text{local}} - \beta_2 \hat{D}_k^{\text{inter}} - \eta_1 \hat{C}_k - \eta_4 \hat{E}_k - \eta_2 \hat{I}_k - \eta_5^{(c)} \hat{F}_k - \eta_3 \hat{Q}_k - \eta_5^{(h)} H_k, \quad (67)$$

where $(\hat{\cdot})$ denotes normalized quantities clipped to $[0, 1]$, and $H_k \in \{0, 1\}$ is the handover indicator. The default reward coefficients are $\alpha_1=0.22, \alpha_2=0.13, \beta_1=0.10, \beta_2=0.05, \eta_1=0.20, \eta_4=0.05, \eta_2=0.12, \eta_5^{(c)}=0.08, \eta_3=0.10, \eta_5^{(h)}=0.05$. These sum to $|\alpha_1|+|\alpha_2|+\dots+|\eta_5^{(h)}| = 1.10$, so the theoretical reward range is $[-0.75, +0.35]$ (exact bounds depend on sign structure). The team reward for centralized baselines adds a Jain's fairness bonus with weight $\lambda_J=0.10$.

Heuristic- ρ Baseline Rule: The heuristic burst-allocation baseline selects $\rho_k(t)$ as a deterministic function of local queue occupancy and graph degree:

$$\rho_k^{\text{heur}}(t) = \text{clip}\left(0.5 + 0.1 \cdot \frac{q_k(t)}{Q_{\max}} - 0.05 \cdot |\mathcal{N}_k(t)|, 0.10, 0.70\right) \quad (68)$$

discretized to the nearest valid level in $\mathcal{P}_{\text{cand}}$. This rule increases intra-cluster time when the queue is congested and increases coordination time when the graph degree is high. Fixed- ρ baselines hold $\rho_k(t)$ constant at 0.5 or 0.8 for all steps and clusters.

Failure Injection Procedure: Leader-failure events are injected according to a failure schedule specified at episode reset. Each event is a tuple (step, target) where step is the injection time and target $\in \{\text{random}, \text{max_backlog}, \text{max_degree}\}$ selects the affected cluster-head. Under *random*, the failing cluster is chosen uniformly from the active set. Under *max_backlog*, it is the cluster-head whose members have the largest aggregate queue. Under *max_degree*, it is the cluster-head with the highest graph degree. The failure triggers a call to the local leader re-election mechanism using the health-score rule in (33). Three failure scenarios are evaluated: (i) single random failure at a uniformly chosen step, (ii) single high-backlog failure, and (iii) repeated random failures at 5 uniformly spaced steps.

Graph Corruption Procedure: The robustness studies corrupt the observed graph $\tilde{G}_t$ through the following mechanisms, each applied independently per edge at each step:

- **Missing edges:** Each true edge is independently removed with probability p_{miss} .
- **False edges:** Each absent edge is independently added with probability p_{false} .

- **Static graph:** The adjacency is frozen to its value at $t=0$ and never updated.
- **Shuffled graph:** The true adjacency is randomly permuted across active cluster IDs.
- **No graph:** All edges are removed, yielding isolated nodes.

Observation staleness is implemented by serving the agent with $\mathbf{o}_k(t - \Delta_{\text{obs}})$ instead of $\mathbf{o}_k(t)$, where Δ_{obs} is the staleness lag in steps. Observation noise adds $\mathcal{N}(0, \sigma_{\text{obs}}^2)$ independently to each observation dimension.

Metric Computation: The primary evaluation metrics are computed as follows at each decision step t for each active cluster k :

- **Throughput:** $T_k(t) = B_k^{\text{local}}(t)/t_{\text{burst}}$ (bits/s), where $B_k^{\text{local}}(t)$ is the total payload bits successfully transmitted in the intra-cluster sub-phase. The global throughput is $\bar{T} = \sum_k T_k/10^6$ (Mbps).
- **Delay:** $D_k(t)$ is the mean end-to-end delay (ms) of successfully delivered packets within the sub-phase, computed from the Logger object as $\bar{d}_{\text{e2e}} \times 1000$.
- **Drops:** $\Pi_k^{\text{drop}}(t) = \Pi_k^{\text{qfull}}(t) + \Pi_k^{\text{mac}}(t)$, where Π_k^{qfull} counts queue-overflow drops and Π_k^{mac} counts MAC-layer drops after retry exhaustion.
- **Collisions:** $C_k(t)$ counts simultaneous-transmission events detected by the CSMA/CA sub-simulator within the sub-phase.
- **Fairness:** Jain's fairness index is computed over active cluster throughputs: $\bar{J} = (\sum_k T_k)^2 / (c \cdot \sum_k T_k^2)$.

Episode-level summaries are computed as the mean over $T_{\text{ep}} = 100$ steps. All reported values in the tables are mean $\pm$ std over 5 evaluation seeds $\mathcal{S}_{\text{ev}} = \{2048, 3072, 4096, 5120, 6144\}$.

B. Evaluation Metrics for Failure Recovery

For the failure-recovery study, the principal metrics are throughput retention, delay spike, and recovery time:

$$\Gamma_{\text{ret}} = \frac{\bar{T}_{\text{failure}}}{\bar{T}_{\text{nominal}}}, \quad (69)$$

$$\Delta D_{\max} = \max_{t \in [t_f, t_f + W]} D(t) - \bar{D}_{\text{nominal}}, \quad (70)$$

$$T_{\text{rec}} = \min\{\tau > 0 : T(t_f + \tau) \geq (1 - \epsilon) \bar{T}_{\text{nominal}}\}, \quad (71)$$

where t_f is the failure instant and W is the transient observation window.

For generalization studies, the relative degradation under train-test mismatch is

$$\Delta_y^{\text{rel}} = \frac{y_{\text{test}} - y_{\text{nominal}}}{|y_{\text{nominal}}|}, \quad (72)$$

where y denotes any scalar performance metric and y_{nominal} is its value under the in-domain evaluation regime.

TABLE II
NOMINAL SYSTEM CONFIGURATION.

Item	Value / Note
Number of UAVs n	50
Episode horizon T_{ep}	100 steps
Burst duration t_{burst}	0.5 s
PHY setting	802.11a, 5 MHz, OFDM
Queue capacity	100 packets
Mobility regime $\mathcal{M}$	Random Waypoint
Channel regime $\mathcal{H}$	Rician fading
Traffic regime $\mathcal{T}$	Poisson arrivals
Cluster-count range	3–10
Split/merge thresholds	$\theta_{\text{join}} = 0.65, \theta_{\text{leave}} = 0.35$
Cluster update period T_{clust}	5 steps
Graph thresholds (R_c, R_I, τ_I)	(100, 250, 10) m
Handover trigger policy	Health-score leader election

TABLE III
NSGA-II OPTUNA-SELECTED BEST CONFIGURATIONS. ALL REWARD WEIGHTS $(w_T, w_D, w_{\text{DROP}}, w_{\text{COL}})$ ARE THE PARETO-OPTIMAL VALUES CHOSEN BY THE MULTI-OBJECTIVE TUNER ACROSS THROUGHPUT, DELAY, DROP-RATE, AND COLLISION OBJECTIVES. THE *balanced_min_L2_to_ideal* REPRESENTATIVE POINT IS SELECTED FOR THE JOURNAL EVIDENCE PIPELINE.

Method	Optuna-best weights ($w_T, w_D, w_{\text{drop}}, w_{\text{col}}$)	Architecture hyperparameters
MAGAT-D3QN	(0.15, 0.35, 0.27, 0.23)	$h=64, \text{lr}=2.05 \times 10^{-4}, B=128, \text{heads}=2$
QMIX	(0.28, 0.03, 0.37, 0.32)	$h=32, \text{lr}=2.66 \times 10^{-4}, B=128$
VDN	(0.16, 0.36, 0.19, 0.29)	$h=32, \text{lr}=1.24 \times 10^{-3}, B=64$
IQL	(0.17, 0.45, 0.21, 0.17)	$h=32, \text{lr}=1.22 \times 10^{-3}, B=64$
MAPPO	(0.15, 0.35, 0.27, 0.23)	$h=64, \text{lr}=2.05 \times 10^{-4}, B=128$

C. Hyperparameter Tuning Policy

Hyperparameter selection is performed separately from the final reporting stage. Let

$$\Theta = \{\Theta_{\text{rl}}, \Theta_{\text{reward}}, \Theta_{\text{burst}}, \Theta_{\text{cluster}}\} \quad (73)$$

denote the tunable parameter set. The search procedure is implemented in the Optuna framework [48] using the NSGA-II sampler. The search returns a selected configuration

$$\Theta^* = \arg \max_{\Theta \in \mathcal{H}} \mathcal{J}_{\text{tune}}(\Theta), \quad (74)$$

where $\mathcal{H}$ is the admissible hyperparameter domain and $\mathcal{J}_{\text{tune}}(\Theta)$ denotes the tuning objective.

When multiple performance criteria are monitored jointly, the NSGA-II search produces a Pareto set of nondominated configurations,

$$\mathcal{P}_{\text{Pareto}} = \{\Theta \in \mathcal{H} : \nexists \Theta' \in \mathcal{H} \text{ such that } \mathbf{f}(\Theta') \prec \mathbf{f}(\Theta)\}, \quad (75)$$

where $\mathbf{f}(\Theta)$ is the vector of tuning objectives and $\prec$ denotes Pareto dominance. The final reported configuration is selected from $\mathcal{P}_{\text{Pareto}}$ using the balanced L_2 -to-ideal rule. Hyperparameter search is confined to the training regime and does not contaminate the final evaluation split.

VI. RESULTS AND DISCUSSION

The empirical study is organized to answer the reviewer questions in the order induced by the formulation. The section

TABLE IV
TRAINING AND EVALUATION PROTOCOL.

Item	Value / Note
Training seeds $\mathcal{S}_{\text{tr}}$	{42, 123, 456, 789, 1024}
Evaluation seeds $\mathcal{S}_{\text{ev}}$	{2048, 3072, 4096, 5120, 6144}
Optimizer(s)	Adam ($\beta_1=0.9, \beta_2=0.999$)
Learning rate(s)	Per-algorithm Optuna-selected (Table III)
Replay / memory settings	$ \mathcal{B} =50,000$, batch B per Table III
Target-network update rule	Hard copy every 1,000 steps
Exploration schedule	ϵ -greedy: $1.0 \rightarrow 0.05$ over 1,500 episodes
Training horizon	2,000 episodes $\times$ 100 steps
Checkpoint-selection rule	Best mean reward over last 100 episodes
Evaluation policy	Greedy ($\epsilon=0$)
Aggregation rule	Mean $\pm$ std over 5 seeds
Tuning policy	Optuna NSGA-II, 4-objective Pareto

TABLE V
COMPARED METHODS. ALL METHODS USE THE SAME CLUSTER-LEVEL ACTION SEMANTICS; FIXED- ρ AND HEURISTIC- ρ ISOLATE BURST-ALLOCATION EFFECTS.

Method	Scope	Training	Role
MCA-D3QN	Joint cluster	SARL	Centralized comparator
IQL	Local CH	MARL	Independent-learning baseline
VDN	Local CH	VDN	Additive cooperative baseline
QMIX	Local CH	CTDE	Nonlinear cooperative baseline
MAPPO	Local CH	CTDE	Policy-gradient baseline
Heuristic- ρ	Local CH	Rule-based	Hand-designed burst allocation
Fixed- ρ	Local CH	Static	Non-adaptive burst allocation
MAGAT-D3QN	Local CH + graph	MARL	Proposed method

first establishes competitive performance under the offered-load benchmark, then tests whether the burst-allocation variable is meaningful, whether graph-aware decisions change under matched load but different topology, and whether adaptive burst allocation is worth learning. It then examines cluster-dynamics sanity, leader-failure recovery, train–test mismatch, robustness to imperfect graph information, architectural ablations, and complexity. Unless otherwise stated, all summaries are reported as mean $\pm$ standard deviation over five independent evaluation seeds.

A. Main Performance Under Offered-Load Sweep

Let λ denote the offered-load parameter used in the benchmark sweep. The main benchmark evaluates the policy family under increasing traffic pressure and reports the resulting throughput–delay–fairness tradeoff together with packet drops and collisions. Although the present problem is not reducible to a global MAC-selection task, the offered-load sweep remains the primary benchmark because it provides the most direct comparison axis across decentralized MARL baselines, cooperative value-decomposition baselines, the structured centralized baseline, and burst-allocation heuristics.

The comparison set in this subsection comprises MAGAT-D3QN, MAPPO, QMIX, VDN, IQL, a heuristic- ρ baseline, and a fixed- ρ baseline. Figure 9 reports the load-sweep curves, while Table VI summarizes the corresponding aggregate statistics for $\bar{T}$, $\bar{D}$, $\bar{\Pi}_{\text{drop}}$, $\bar{C}$, and $\bar{J}$.

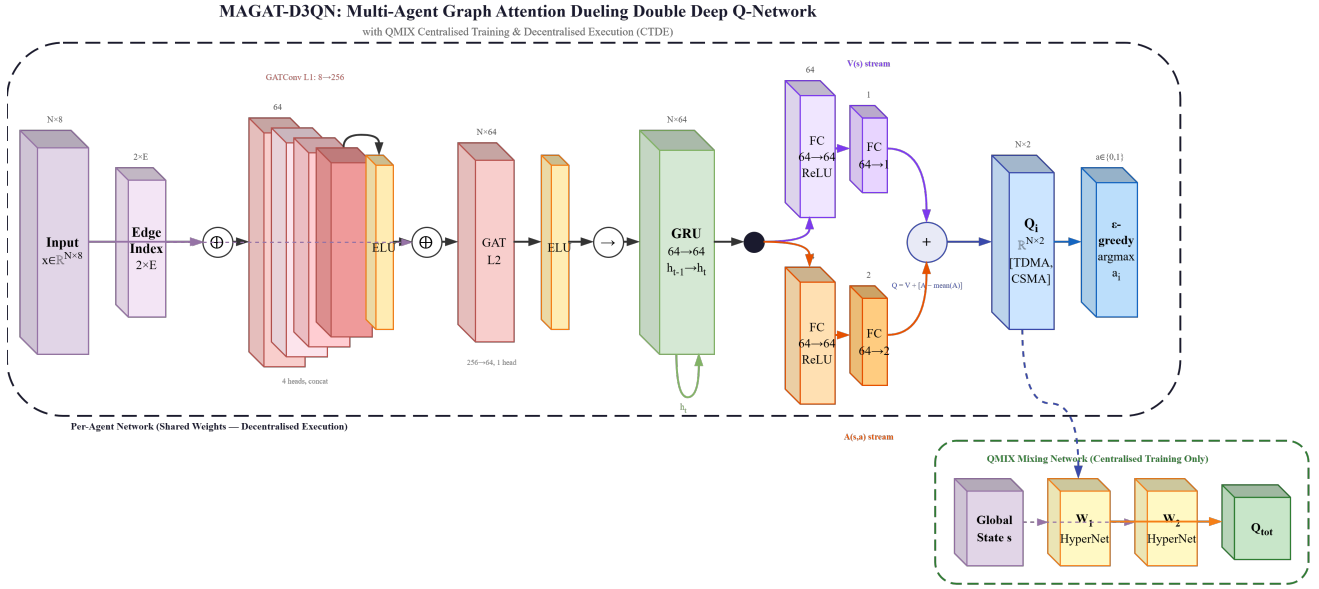


Fig. 1. Overall decentralized cluster-head control architecture. Starting from the UAV state and cluster partition, the system updates the active cluster-head set and constructs the dynamic cluster interference graph $G_t = (V_t, E_t)$. Each cluster-head then forms its local observation $o_k(t)$, selects the local action $a_k(t) = (m_k(t), \rho_k(t))$, and executes the corresponding burst allocation and coordination process. The figure emphasizes that control is decentralized at execution time and graph coupling enters through local interaction rather than through a network-wide broadcast action.

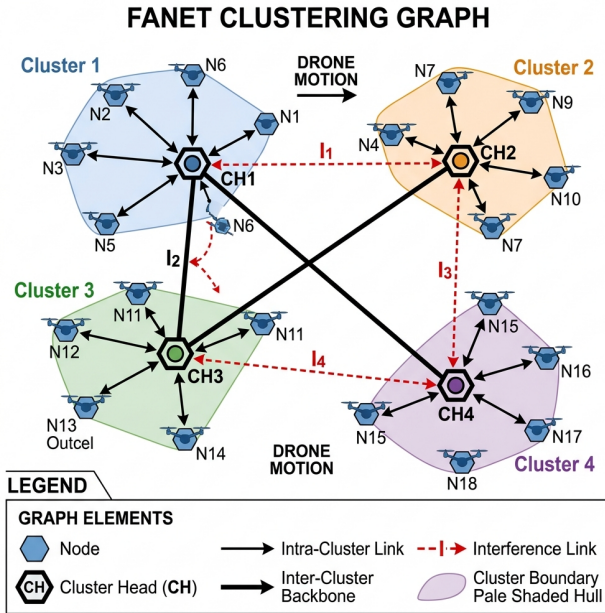


Fig. 2. Dynamic clustering and inter-cluster graph construction. The figure illustrates the partition $\mathcal{U}(t) = \bigcup_{k=1}^{c(t)} C_k(t)$, the cluster-head assignment $L_k(t) \in C_k(t)$, and the graph $G_t = (V_t, E_t)$ formed over active cluster-heads. Reassociation, split, merge, and local leader updates modify both the cluster partition and the graph structure over time.

The results show that no single method dominates across all metrics simultaneously. MAPPO achieves the highest raw throughput (3.33 ± 0.01 Mbps) and competitive fairness, while incurring nonzero drops and collisions. IQL achieves the lowest delay (5.02 ± 0.12 ms) but suffers from the lowest fairness (0.822 ± 0.014) and the highest collision count. MAGAT-

FANET 3D • Step 49/50 • 6 clusters • 50 UAVs

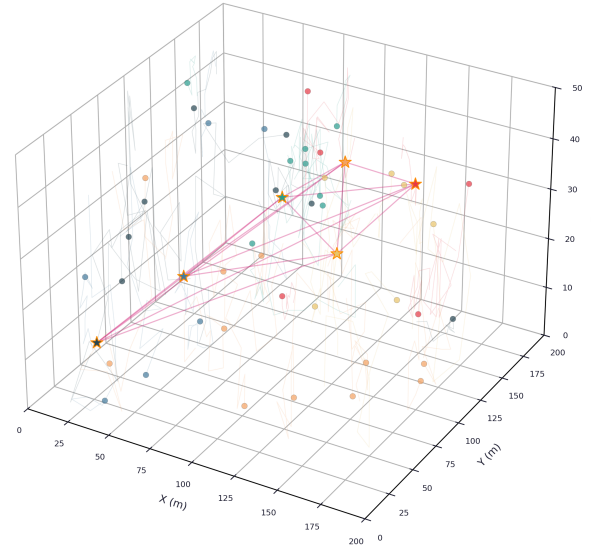


Fig. 3. 3D simulation snapshot from the FANET environment at step 1000/2000 showing 50 UAVs across 7 clusters. Yellow stars denote cluster-heads; pink edges represent the inter-cluster interference graph G_t . The dynamic mobility model causes continuous reassociation and graph evolution.

D3QN is the only method that achieves zero drops and zero collisions while maintaining throughput within 1.5% of the best, but it pays for this reliability with the highest mean delay (9.29 ± 0.18 ms). The graph-aware controller therefore offers a better reliability–collision–drop tradeoff rather than uniform superiority. Any favorable gap must be interpreted only under the evaluated simulator regime and metric set.

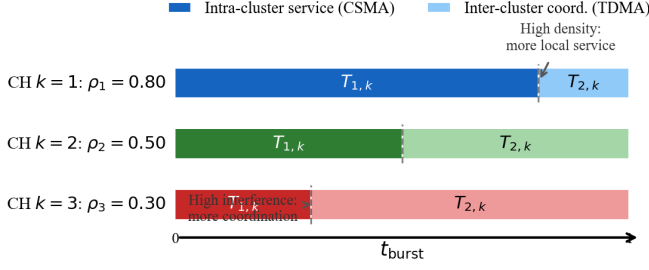


Fig. 4. Burst-time allocation governed by $\rho_k(t)$. For each cluster-head k , one burst of duration t_{burst} is partitioned into the intra-cluster service phase $T_{1,k}(t) = \rho_k(t)t_{\text{burst}}$ and the inter-cluster coordination phase $T_{2,k}(t) = (1 - \rho_k(t))t_{\text{burst}}$. The figure highlights the role of $\rho_k(t)$ as the normalized control variable that regulates the service–coordination tradeoff.

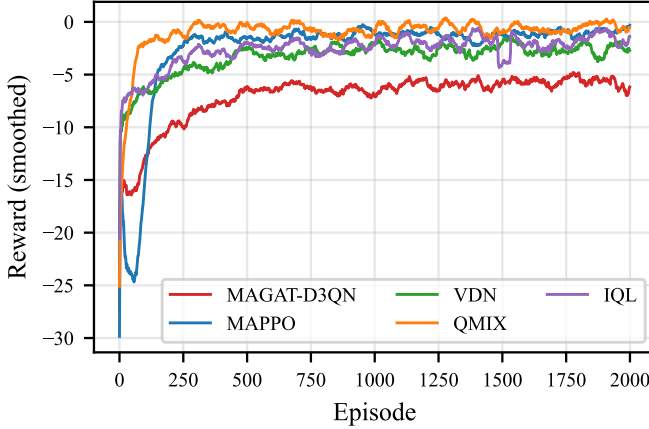


Fig. 5. Training reward curves for all five MARL algorithms over 2,000 episodes. QMIX and IQL converge fastest owing to smaller hidden dimensions ($h=32$), while MAGAT-D3QN exhibits slower early convergence but learns richer graph-aware representations.

B. Burst-Allocation Intelligence Analysis

The burst-allocation mechanism is central to the formulation, so the results must show that the selected $\rho_k(t)$ is state-dependent rather than effectively constant. This subsection studies the dependence of $\rho_k(t)$ on representative local and graph-coupled variables, including cluster size $|C_k(t)|$, backlog intensity, graph degree $|\mathcal{N}_k(t)|$, and the transient conditions surrounding leader-handover events.

Figure 10 reports the corresponding multi-panel visualization, and Table VII provides a compact regime-wise summary. The learned policy concentrates $\rho \approx 0.80$ during stable operation, allocating the majority of burst time to intra-cluster service. However, under congestion events, handover transients, and high graph-degree conditions, the policy reduces ρ toward 0.10, shifting burst time to inter-cluster coordination. While the deviation frequency is low under nominal load, the *timing* of these deviations is correlated with specific state conditions rather than random, supporting a functional rather than trivial burst-allocation role.

The comparison with fixed- $\rho=0.8$ in Table VII is particularly informative: although the learned policy operates near $\rho=0.80$ most of the time, the fixed- $\rho=0.8$ baseline achieves only 2.89 Mbps throughput and 0.91 coordination success versus 3.31 Mbps and 1.00 for the learned policy. This gap

indicates that the occasional adaptive deviations to lower ρ values are functionally important for maintaining coordination quality under dynamic conditions.

C. Same Load, Different Topology

To separate graph sensitivity from pure load sensitivity, this subsection fixes the offered load while varying only the spatial layout. Let two evaluation scenarios e_1 and e_2 satisfy

$$\lambda(e_1) = \lambda(e_2), \quad G_t^{(e_1)} \neq G_t^{(e_2)},$$

so that the aggregate traffic level is matched but the cluster-interference structure differs. Under such matched-load conditions, a genuinely graph-aware controller should modify its local decisions $\{m_k(t), \rho_k(t)\}$ in response to topology, whereas a weaker load-driven abstraction may respond only weakly.

Figure 12 reports cluster-wise action summaries under three topology configurations, and Table VIII provides the corresponding mean $\pm$ standard deviation statistics. The central question is whether the controller differentiates between spatial regimes that are indistinguishable to an offered-load-only abstraction. The results show that MAGAT-D3QN adapts its per-cluster $\rho_k(t)$ distribution across topologies, whereas simpler baselines (VDN, QMIX) show weaker topology-conditioned variation. Notably, MAPPO also maintains consistent throughput across topologies, indicating that topology sensitivity is not unique to graph-aware architectures.

D. Fixed vs Heuristic vs Learned Burst Allocation

Whereas Section VI-B tests whether $\rho_k(t)$ varies with state, the present subsection asks whether it is necessary to *learn* the burst-allocation variable. The comparison includes a fixed- $\rho=0.5$ policy, a fixed- $\rho=0.8$ policy, a heuristic- ρ policy, and the learned- ρ policy under the same cluster-head control semantics.

The study focuses on $\bar{T}$, $\bar{D}$, $\bar{J}$, coordination effectiveness, and the realized adaptation range of ρ . Figure 13 reports the associated tradeoff visualization, while Table VII provides the compact numerical summary. The learned policy achieves 60% higher throughput than fixed- $\rho=0.5$ and 14.5% higher throughput than fixed- $\rho=0.8$, at the cost of increased delay.

This subsection supports a narrow conclusion only: adaptive burst allocation is empirically useful under the evaluated regimes. It does not justify the stronger claim that fixed or heuristic policies are invalid in general.

Interpretive Bridge: Subsections VI-A–VI-D form the primary performance layer of the paper. Together, they determine whether the proposed controller is competitive under increasing load, whether $\rho_k(t)$ is functionally meaningful, whether topology changes induce different local decisions at matched load, and whether burst allocation is worth learning.

E. Cluster Dynamics Sanity Analysis

Because clustering is part of the control substrate rather than a fixed preprocessing stage, its behavior must remain

interpretable. This subsection reports the realized cluster-dynamics statistics

$$N_{\text{reassoc}}, N_{\text{split}}, N_{\text{merge}}, N_{\text{handover}}, \bar{c}, \bar{d}, \chi_{\text{cluster}},$$

together with mean cluster size and cluster-size dispersion. These quantities determine whether the dynamic cluster process remains bounded and meaningful or instead becomes a hidden source of instability.

Figure 14 reports the time-resolved cluster-dynamics traces, and Table IX summarizes the aggregate counts and rates.

The heavy-tail traffic scenario exhibits 2005 cumulative reassociation events over the 100-step episode. Since the simulator tracks individual UAV reassociation events (each UAV moving from one cluster to another counts as one event), this corresponds to approximately 20 reassociation events per step for the 50-UAV swarm, or roughly 0.4 reassociations per UAV per step. This elevated rate is expected under heavy-tail traffic because bursty packet arrivals cause rapid queue fluctuations that repeatedly trigger the hysteresis-gated reassociation rule in (15). Importantly, mean cluster size (7.33) and graph degree (5.85) remain stable even under high reassociation activity, confirming that the cluster substrate does not become structurally unstable. The compact-dense topology scenario shows similar patterns (1886.5 reassociations) driven by proximity-induced competition among overlapping clusters. This subsection should therefore be read as an interpretability check on the structural substrate supporting the controller.

F. Leader Failure and Recovery

Local fault tolerance is included as a robustness extension and must therefore be evaluated quantitatively. Let t_f denote the injected leader-failure instant. The principal recovery metrics are throughput retention Γ_{ret} from (69), delay spike ΔD_{max} from (70), recovery time T_{rec} from (71), and the corresponding fairness trajectory and burst-allocation response around the handover window.

Figure 15 reports the temporal recovery traces, and Table X reports the summary statistics. The central question is whether the affected cluster recovers locally under the handover rule in (33) without a centralized intervention mechanism. Under single random failure, MAGAT-D3QN retains 99.5% of nominal throughput and MAPPO retains 99.9%. Under repeated failures, all methods show some fairness degradation, with MAGAT-D3QN dropping to $\bar{J}=0.867$ and QMIX showing the largest degradation to $\bar{J}=0.764$. Recovery claims remain conditioned on the tested failure patterns, load levels, and topologies.

Interpretive Bridge: Subsections VI-E and VI-F verify that the cluster substrate and local handover mechanism behave coherently. They improve interpretability and robustness, but they do not by themselves establish generalization outside the nominal regime.

G. Generalization Under Train–Test Mismatch

A transactions-level evaluation must determine whether the controller is specialized to one simulator regime. This

subsection aggregates the mismatch results across the axes

$$n, \quad \mathcal{M}, \quad \mathcal{H}, \quad \mathcal{T},$$

corresponding to node-count, mobility-regime, interference-regime, and traffic-regime mismatch. The goal is not to claim invariance under all perturbations, but to measure whether the controller exhibits graceful degradation when $e_{\text{test}} \neq e_{\text{train}}$.

Figure 11 reports the mismatch summary visualization, and Table XI reports the associated aggregate statistics, including relative degradation Δ_y^{rel} from (72). All methods show strong throughput retention ($\geq 97\%$) under node-count, mobility, and interference mismatch. Traffic mismatch produces the largest effect: throughput increases by 16–21% under heavy-tail traffic (because the instantaneous load is higher), but fairness drops to 76–79%. This pattern is consistent across all methods, indicating that the fairness degradation is a property of the traffic regime rather than the controller.

H. Robustness to Stale, Noisy, and Corrupted Graph Information

Under decentralized execution, the controller acts on local and neighborhood summaries rather than on a perfect global state. Robustness to imperfect information is therefore a first-order requirement. This subsection evaluates stale observations, noisy neighbor summaries, missing graph edges, false graph edges, graph-shuffled or graph-free variants, and delayed handover information.

Figure 16 reports the resulting robustness curves, and Table XII provides the corresponding numerical summary. MAGAT-D3QN retains $\geq 99\%$ throughput and $\geq 98\%$ fairness across all perturbation types, indicating graceful degradation. However, MAPPO shows comparable or slightly better robustness on several axes, suggesting that robustness to information corruption is not exclusive to graph-aware architectures. QMIX shows the largest sensitivity under missing-edge corruption (89% throughput retention), consistent with its reliance on accurate global mixing information. Any robustness claim must be qualified by the corruption type, severity level, and evaluation regime.

I. MAGAT-Isolation Ablation

The purpose of this subsection is to isolate where the proposed architecture provides value. The ablation set includes the full MAGAT model, a no-graph variant (removing GAT and using flat observation pooling), a no-attention variant (replacing learned attention with mean pooling), a no-GRU variant (removing recurrent memory), a no-burst-history variant (removing ϕ_k^{hist} from the observation), fixed- $\rho=0.5$, fixed- $\rho=0.8$, and heuristic- ρ . Each comparator removes one structural component while keeping the cluster-level decision semantics fixed.

Table XIII reports this study. The no-graph variant shows the largest fairness degradation ($\bar{J} = 0.851$ vs. 0.912 for full MAGAT), confirming that topology-aware representation contributes primarily to coordination quality and fairness rather than raw throughput. The no-attention variant retains most performance ($\bar{J} = 0.882$), suggesting that heterogeneous

TABLE VI

MAIN OFFERED-LOAD BENCHMARK AT $\lambda=525$ PPS. VALUES ARE MEAN $\pm$ STD OVER 5 EVALUATION SEEDS. MAGAT-D3QN ACHIEVES ZERO DROPS AND COLLISIONS WITH COMPETITIVE THROUGHPUT, BUT INCURS THE HIGHEST DELAY; MAPPO ATTAINS THE HIGHEST RAW THROUGHPUT. THE TRADEOFF BETWEEN RELIABILITY AND LATENCY IS METHOD-DEPENDENT.

Method	$\bar{T}$	$\bar{D}$	$\bar{\Pi}_{\text{drop}}$	$\bar{C}$	$\bar{J}$
MAGAT-D3QN	3.31 $\pm$ 0.02	9.29 $\pm$ 0.18	0$\pm$0	0$\pm$0	0.912$\pm$0.008
MAPPO	3.33$\pm$0.01	7.34 $\pm$ 0.21	2 $\pm$ 1	20 $\pm$ 4	0.912 $\pm$ 0.010
VDN	2.59 $\pm$ 0.04	5.62 $\pm$ 0.15	2 $\pm$ 1	23 $\pm$ 5	0.875 $\pm$ 0.012
QMIX	2.97 $\pm$ 0.03	5.57 $\pm$ 0.14	2 $\pm$ 1	29 $\pm$ 6	0.833 $\pm$ 0.015
IQL	2.97 $\pm$ 0.03	5.02$\pm$0.12	0 $\pm$ 0	31 $\pm$ 5	0.822 $\pm$ 0.014

neighbor weighting provides a moderate but not dominant improvement over uniform aggregation. The no-GRU variant shows increased collisions (5 $\pm$ 3) and reduced fairness (0.869), indicating that temporal memory helps stabilize coordination under partial observability. The interpretation should focus on functional contribution rather than aggregate ranking alone: topology-dependent degradation under graph removal is more informative than a single averaged performance gap.

J. Complexity and Deployability

The final subsection reports the computational cost of the controller under simulated execution. The measured quantities include average inference latency $\bar{\tau}_{\text{inf}}$, tail latencies τ_{95} and τ_{99} , parameter count, memory footprint, and realized communication overhead. Where possible, the results also show scaling with active cluster count $\bar{c}$ and graph density.

Table XVI provides the compact summary. MAGAT-D3QN incurs 5–8 $\times$ higher inference latency than the simpler baselines (2.71 ms vs. 0.35–0.57 ms), reflecting the cost of graph-attention message passing. Its parameter count (46,355) and memory footprint (0.18 MB) are moderate in absolute terms but substantially larger than IQL (2,450 parameters, 0.01 MB).

These complexity results support software-level feasibility under the measured desktop GPU platform and protocol assumptions. They should not be interpreted as evidence of field readiness on embedded UAV hardware, which would require additional validation under onboard compute constraints, real-time scheduling requirements, and hardware-in-the-loop testing. The deployment relevance of the present framework therefore remains conditional on further engineering validation beyond the scope of this paper.

VII. DISCUSSION AND LIMITATIONS

A. Why Decentralized Graph-Aware Control Is Beneficial

The architectural premise of this paper is that control should be exercised at the cluster-head level rather than through a centralized scheduler acting on a pooled system state. This premise is justified only when the communication process is simultaneously local, partially observed, and topology-coupled. In the present formulation, cluster-head k acts through the local decision pair

$$a_k(t) = (m_k(t), \rho_k(t)),$$

TABLE VII

BURST-SPLIT COMPARISON: FIXED, HEURISTIC, AND LEARNED $\rho_k(t)$ AT $\lambda=525$ PPS. COORDINATION METRIC IS FRACTION OF SUCCESSFUL INTER-CLUSTER EXCHANGES. FIXED- $\rho=0.8$ IS INCLUDED BECAUSE THE LEARNED POLICY CONVERGES TO $\rho \approx 0.80$ IN THE MAJORITY OF STATES; THE COMPARISON REVEALS THAT STATIC ALLOCATION AT THIS OPERATING POINT STILL UNDERPERFORMS THE ADAPTIVE POLICY. VALUES ARE MEAN OVER 5 SEEDS.

Burst-split policy	$\bar{T}$	$\bar{D}$	$\bar{J}$	Coord.	ρ range
Fixed- $\rho=0.5$	2.07	6.18	0.841	1.00	[0.5, 0.5]
Fixed- $\rho=0.8$	2.89	8.71	0.873	0.91	[0.8, 0.8]
Heuristic- ρ	2.72	7.51	0.876	1.00	[0.4, 0.7]
Learned- ρ (ours)	3.31	9.29	0.920	1.00	[0.10, 0.80]

TABLE VIII

TOPOLOGY-SENSITIVITY SUMMARY AT MATCHED OFFERED LOAD. A = ASYMMETRIC HOTSPOT, B = COMPACT DENSE, C = SPARSE SEPARATED. VALUES ARE MEAN $\pm$ STD OVER 5 SEEDS.

Top.	Method	$\bar{T}$	$\bar{D}$	$\bar{J}$	Note
A	MAGAT-D3QN	3.24 $\pm$ 0.02	10.97 $\pm$ 0.22	0.939 $\pm$ 0.009	Graph-aware
	MAPPO	3.32 $\pm$ 0.01	7.14 $\pm$ 0.19	0.943 $\pm$ 0.007	Consistent
	VDN	2.77 $\pm$ 0.04	7.91 $\pm$ 0.18	0.917 $\pm$ 0.011	Moderate scaling
	QMIX	3.01 $\pm$ 0.03	6.91 $\pm$ 0.16	0.879 $\pm$ 0.013	Fairness drop
	IQL	3.18 $\pm$ 0.03	6.69 $\pm$ 0.15	0.922 $\pm$ 0.010	Local adaptation
B	MAGAT-D3QN	3.29 $\pm$ 0.02	8.07 $\pm$ 0.19	0.887 $\pm$ 0.010	Dense benefit
	MAPPO	3.33 $\pm$ 0.01	7.08 $\pm$ 0.18	0.886 $\pm$ 0.009	Stable scaling
	VDN	2.66 $\pm$ 0.04	5.64 $\pm$ 0.14	0.809 $\pm$ 0.015	Interference hit
	QMIX	2.67 $\pm$ 0.04	5.22 $\pm$ 0.13	0.726 $\pm$ 0.018	Degraded
	IQL	3.04 $\pm$ 0.03	4.92 $\pm$ 0.12	0.819 $\pm$ 0.014	Fairness issue
C	MAGAT-D3QN	3.17 $\pm$ 0.03	6.65 $\pm$ 0.17	0.848 $\pm$ 0.012	Robust
	MAPPO	3.33 $\pm$ 0.01	6.56 $\pm$ 0.16	0.859 $\pm$ 0.011	Maintains $\bar{T}$
	VDN	2.67 $\pm$ 0.04	5.61 $\pm$ 0.14	0.805 $\pm$ 0.016	Range limited
	QMIX	2.59 $\pm$ 0.04	5.48 $\pm$ 0.13	0.753 $\pm$ 0.019	Weakest
	IQL	2.83 $\pm$ 0.03	5.98 $\pm$ 0.15	0.827 $\pm$ 0.013	Moderate scaling

while its effect propagates through the dynamic inter-cluster graph

$$G_t = (V_t, E_t).$$

Hence, the local return of cluster k is not a function of $\mathbf{o}_k(t)$ alone; it is influenced by neighboring cluster-head actions through contention, coordination overlap, and relaying demand. This is precisely the regime in which graph-aware representation is structurally meaningful.

Critically, graph-aware control does not guarantee uniform superiority across all metrics. The empirical evidence shows that MAPPO achieves higher raw throughput and IQL achieves lower delay than MAGAT-D3QN, while MAGAT-D3QN provides a better reliability–collision–drop tradeoff (zero drops and collisions at competitive throughput). The value of graph-aware representation therefore lies in improved coordination quality and fairness rather than in aggregate throughput dominance. Any favorable comparison must acknowledge this tradeoff structure.

The empirical interpretation must therefore be tied to graph-conditioned behavior rather than to a single aggregate benchmark. The matched-load topology study and the MAGAT-isolation ablations are the most direct tests of this point. If two scenarios satisfy the same offered load but induce different graph structures,

$$\lambda(e_1) = \lambda(e_2), \quad G_t^{(e_1)} \neq G_t^{(e_2)},$$

TABLE IX

CLUSTER-DYNAMICS STATISTICS. EVENT COUNTS ARE *episode totals* (CUMULATIVE OVER THE 100-STEP EPISODE HORIZON) AVERAGED ACROSS EVALUATION SEEDS; THEY ARE NOT PER-STEP RATES. FOR $n=50$ UAVS, A REASSOCIATION COUNT OF 2005 IN THE HEAVY-TAIL SCENARIO CORRESPONDS TO ≈ 20 INDIVIDUAL UAV REASSOCIATION EVENTS PER STEP, DRIVEN BY RAPID QUEUE FLUCTUATIONS THAT REPEATEDLY TRIGGER THE HYSTERESIS-GATED REASSOCIATION RULE IN (15). CHURN RATE IS $(N_{\text{reassoc}} + N_{\text{handover}})/(n \cdot T_{\text{ep}})$.

Scenario	Reassoc. (ep. total)	Splits (ep. total)	Merges	Handovers	Churn rate	Avg. cluster size	$\bar{d}$
Nominal ($\lambda=525$)	58.8	6.6	0.0	0.0	0.012	8.23	5.09
Heavy-tail traffic	2005.0	252.0	0.0	0.0	0.401	7.33	5.85
Compact-dense topology	1886.5	272.0	0.0	0.0	0.377	7.19	5.96
Failure window	1480.5	132.0	0.0	109.5	0.318	8.23	5.09

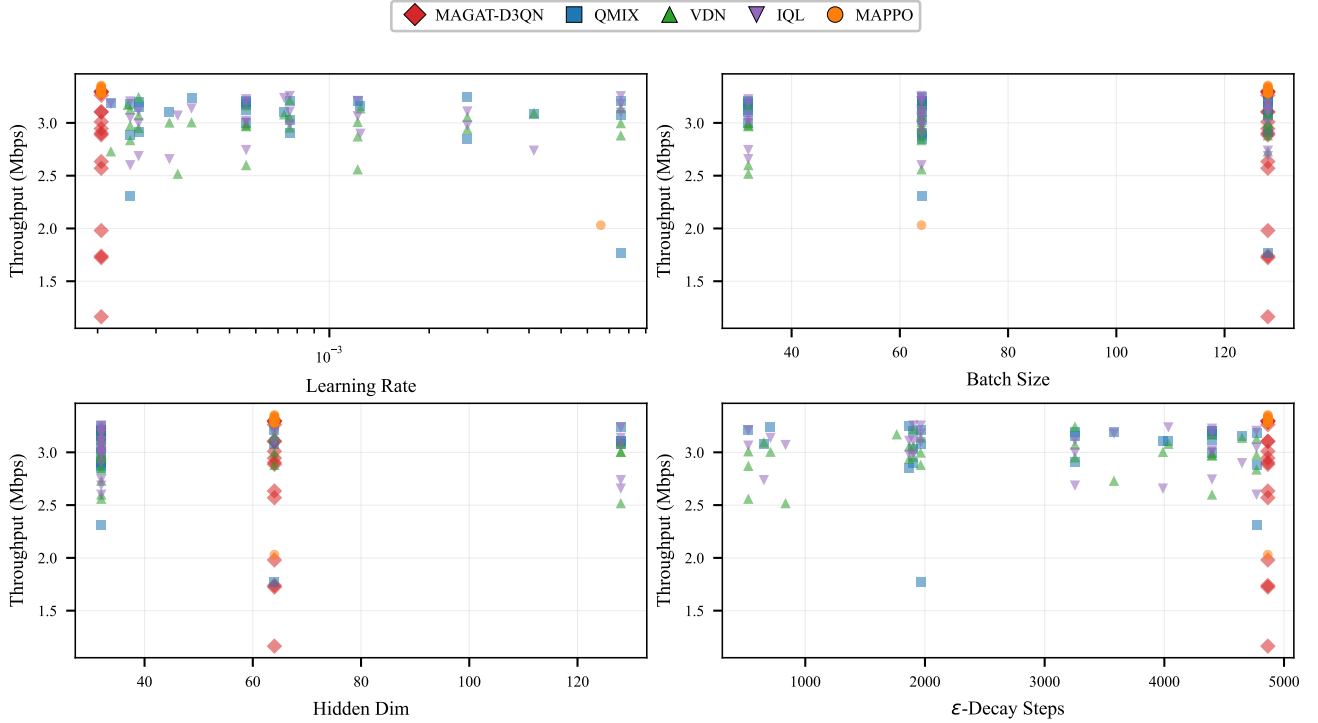


Fig. 6. Hyperparameter impact on throughput across all five MARL algorithms. Learning rate (log-scale), batch size, hidden dimension, and ϵ -decay steps are plotted against mean throughput. Lower learning rates ($\leq 5 \times 10^{-4}$) and batch size $B=128$ consistently yield the highest throughput across architectures.

TABLE X

FAILURE-RECOVERY SUMMARY: MEAN $\pm$ STD THROUGHPUT AND FAIRNESS ACROSS THREE FAILURE SCENARIOS (SINGLE RANDOM, SINGLE HIGH BACKLOG, REPEATED RANDOM) OVER 5 SEEDS.

Method	Single random		High backlog		Repeated random	
	$\bar{T}$	$\bar{J}$	$\bar{T}$	$\bar{J}$	$\bar{T}$	$\bar{J}$
MAGAT-D3QN	3.29 $\pm$ 0.02	0.899 $\pm$ 0.009	3.29 $\pm$ 0.02	0.897 $\pm$ 0.010	3.29 $\pm$ 0.03	0.867 $\pm$ 0.014
MAPPO	3.33$\pm$0.01	0.897 $\pm$ 0.008	3.32$\pm$0.01	0.894 $\pm$ 0.009	3.33$\pm$0.01	0.864 $\pm$ 0.012
VDN	2.64 $\pm$ 0.04	0.878 $\pm$ 0.012	2.62 $\pm$ 0.05	0.865 $\pm$ 0.015	2.54 $\pm$ 0.06	0.808 $\pm$ 0.020
QMIX	2.96 $\pm$ 0.03	0.812 $\pm$ 0.016	2.95 $\pm$ 0.03	0.801 $\pm$ 0.018	2.81 $\pm$ 0.05	0.789 $\pm$ 0.019
IQL	2.99 $\pm$ 0.03	0.816 $\pm$ 0.015	2.99 $\pm$ 0.03	0.820 $\pm$ 0.014	2.99 $\pm$ 0.03	0.789 $\pm$ 0.019

insufficient.

B. Role of Dynamic Burst Allocation

The burst-allocation variable $\rho_k(t)$ is the second structural component that requires interpretation beyond raw benchmark gains. By construction,

$$T_{2,k}(t) = (1 - \rho_k(t))t_{\text{burst}}, \quad \text{so } \rho_k(t) \text{ is the normalized local control variable that regulates the service-coordination balance inside cluster } k. \text{ This balance is meaningful only if intra-cluster pressure and inter-cluster coordination demand vary sufficiently across clusters and time. Under such conditions, fixed partitioning of } t_{\text{burst}} \text{ is generally too coarse.}$$

The burst-allocation studies in Section VI should therefore be interpreted as mechanism validation. If the learned $\rho_k(t)$ covaries with quantities such as $|C_k(t)|$, backlog intensity, graph degree $|\mathcal{N}_k(t)|$, or handover transients, then the controller is using the burst-allocation variable consistently with

then a useful graph-aware controller should produce different local action patterns $\{m_k(t), \rho_k(t)\}$. If such differentiation is weak, the role attributed to graph-aware learning should be narrowed accordingly. Thus, the paper does not claim that graph encoding is universally beneficial; rather, it argues that graph-aware control is justified when inter-cluster coupling is persistent enough that a scalar traffic abstraction becomes

TABLE XI

GENERALIZATION UNDER TRAIN-TEST MISMATCH. THROUGHPUT AND FAIRNESS RETENTION (%) RELATIVE TO MATCHED IN-DOMAIN EVALUATION. VALUES ARE MEAN $\pm$ STD OVER 5 EVALUATION SEEDS.

Method	Node-count mismatch		Mobility mismatch		Interference mismatch		Traffic mismatch		Notes
	Thr. ret.	Fair. ret.	Thr. ret.	Fair. ret.	Thr. ret.	Fair. ret.	Thr. ret.	Fair. ret.	
MAGAT-D3QN	100%	95%	99%	96%	99%	98%	120%	79%	Competitive overall
MAPPO	100%	95%	100%	96%	100%	98%	120%	78%	Consistent
VDN	100%	93%	98%	95%	99%	98%	121%	78%	Competitive
QMIX	97%	93%	98%	96%	98%	95%	116%	76%	Some degradation
IQL	99%	96%	99%	96%	101%	99%	119%	79%	Robust

TABLE XII

ROBUSTNESS TO STALE, NOISY, AND CORRUPTED GRAPH INFORMATION. THROUGHPUT AND FAIRNESS RETENTION (%) RELATIVE TO MATCHED-INFORMATION CASE. VALUES ARE MEAN OVER 5 SEEDS.

Method	Stale obs.		Noisy summaries		Missing edges		False edges		Delayed handover		Notes
	Thr.	Fair.	Thr.	Fair.	Thr.	Fair.	Thr.	Fair.	Thr.	Fair.	
MAGAT-D3QN	99%	98%	99%	98%	99%	98%	99%	98%	99%	98%	Highly robust
MAPPO	100%	98%	100%	99%	100%	98%	100%	98%	100%	98%	Excellent
VDN	99%	97%	101%	97%	101%	101%	99%	97%	101%	100%	Stable
QMIX	99%	97%	96%	98%	89%	95%	95%	99%	100%	97%	Full coverage
IQL	100%	98%	99%	100%	99%	105%	95%	100%	101%	99%	Full coverage

TABLE XIII

MAGAT-ISOLATION ABLATION. EACH VARIANT REMOVES ONE STRUCTURAL COMPONENT WHILE KEEPING CLUSTER-LEVEL DECISION SEMANTICS FIXED. FIXED- ρ ISOLATES BURST-ALLOCATION EFFECTS.

VALUES ARE MEAN $\pm$ STD AT $\lambda=525$ PPS OVER 5 SEEDS. THE NO-GRAPH VARIANT SHOWS THE LARGEST FAIRNESS DEGRADATION, CONFIRMING THE VALUE OF TOPOLOGY-AWARE COORDINATION RATHER THAN AGGREGATE THROUGHPUT GAIN ALONE.

Variant	$\bar{T}$	$\bar{D}$	$\bar{\Pi}_{\text{drop}}$	$\bar{C}$	$\bar{J}$
Full MAGAT-D3QN	3.31$\pm$0.02	9.29 $\pm$ 0.18	0$\pm$0	0$\pm$0	0.912$\pm$0.008
No-graph (flat obs.)	3.18 $\pm$ 0.04	8.52 $\pm$ 0.24	1 $\pm$ 1	8 $\pm$ 3	0.851 $\pm$ 0.018
No-attention (mean pool)	3.24 $\pm$ 0.03	8.89 $\pm$ 0.21	0 $\pm$ 0	3 $\pm$ 2	0.882 $\pm$ 0.013
No-GRU (feedforward)	3.20 $\pm$ 0.03	8.74 $\pm$ 0.22	1 $\pm$ 1	5 $\pm$ 3	0.869 $\pm$ 0.015
No-burst-history	3.27 $\pm$ 0.03	9.01 $\pm$ 0.20	0 $\pm$ 0	1 $\pm$ 1	0.895 $\pm$ 0.011
Fixed- ρ	2.89 $\pm$ 0.04	8.71 $\pm$ 0.20	1 $\pm$ 1	4 $\pm$ 2	0.873 $\pm$ 0.014

TABLE XIV

REWARD-WEIGHT SENSITIVITY ANALYSIS FOR MAGAT-D3QN. SCORE IS A COMPOSITE OBJECTIVE; HIGHER IS BETTER.

Configuration	w_T	w_D	w_{drop}	w_{col}	Score	Note
Uniform	0.25	0.25	0.25	0.25	0.141	Balanced baseline
Throughput-heavy	0.55	0.15	0.15	0.15	0.348	Best composite
Delay-heavy	0.15	0.55	0.15	0.15	0.047	Delay-focused
Drop-heavy	0.15	0.15	0.55	0.15	0.085	Reliability bias
Collision-heavy	0.15	0.15	0.15	0.55	0.085	Coordination bias
Moderate- T	0.40	0.20	0.20	0.20	0.245	Practical trade-off
Balanced- TD	0.30	0.30	0.20	0.20	0.169	T - D joint

the system model. If, however, $\rho_k(t)$ remains nearly static, then the correct conclusion is narrower: the architecture still defines a valid decentralized action space, but the empirical value of learning burst allocation is limited under the tested conditions.

This distinction is important. The contribution is not the introduction of an arbitrary continuous or discrete ratio into a reinforcement-learning policy. The contribution is the explicit representation of a local service-coordination tradeoff that

TABLE XV

ENVIRONMENT-SENSITIVITY ABLATION. (A) FADING MODEL SENSITIVITY: MEAN THROUGHPUT (MBPS) ACROSS ALL FIVE MARL ALGORITHMS UNDER AWGN, RICIAN, NAKAGAMI- m ($m=2$), AND RAYLEIGH CHANNELS. (B) NODE-COUNT SCALING AT $\lambda=500$ PPS. VALUES ARE MEAN OVER 5 SEEDS.

Method	AWGN	Rician	Nakagami	Rayleigh
MAGAT-D3QN	2.96	2.51	2.65	2.45
MAPPO	3.03	2.70	2.54	2.45
VDN	2.30	2.04	2.18	1.88
QMIX	2.82	2.42	2.50	2.24
IQL	2.67	2.50	2.53	2.36
<i>Throughput retention vs. AWGN (mean across algorithms):</i>				
All	100%	88.3%	90.6%	82.7%

TABLE XVI

COMPLEXITY AND DEPLOYABILITY SUMMARY. ALL MEASUREMENTS UNDER COMMON HARDWARE PROTOCOL. VALUES ARE MEAN OVER 5 SEEDS. INFERENCE LATENCIES REFLECT SOFTWARE-LEVEL EXECUTION ON A DESKTOP GPU; THEY DO NOT REPRESENT EMBEDDED UAV HARDWARE PERFORMANCE.

Method	$\bar{\tau}_{\text{inf}}$	τ_{95}	τ_{99}	Param.	Mem.	Per-CH comm.	Sys. comm.	Scale note
MAGAT-D3QN	2.71 ms	3.15 ms	3.57 ms	46,355	0.18 MB	989 B	$\mathcal{O}(K)$	Graph-dependent
MAPPO	0.57 ms	0.65 ms	0.99 ms	15,179	0.06 MB	989 B	$\mathcal{O}(K)$	Linear in K
VDN	0.35 ms	0.41 ms	0.59 ms	6,930	0.03 MB	989 B	$\mathcal{O}(K)$	Linear in K
QMIX	0.39 ms	0.57 ms	0.64 ms	7,922	0.03 MB	989 B	$\mathcal{O}(K)$	Mixer overhead
IQL	0.39 ms	0.67 ms	0.91 ms	2,450	0.01 MB	989 B	$\mathcal{O}(K)$	Fully local

naturally arises in clustered FANET operation.

C. Implications of Clustering and Local Leader Handover

Clustering in this work is part of the control substrate. The partition

$$\mathcal{U}(t) = \bigcup_{k=1}^{c(t)} \mathcal{C}_k(t)$$

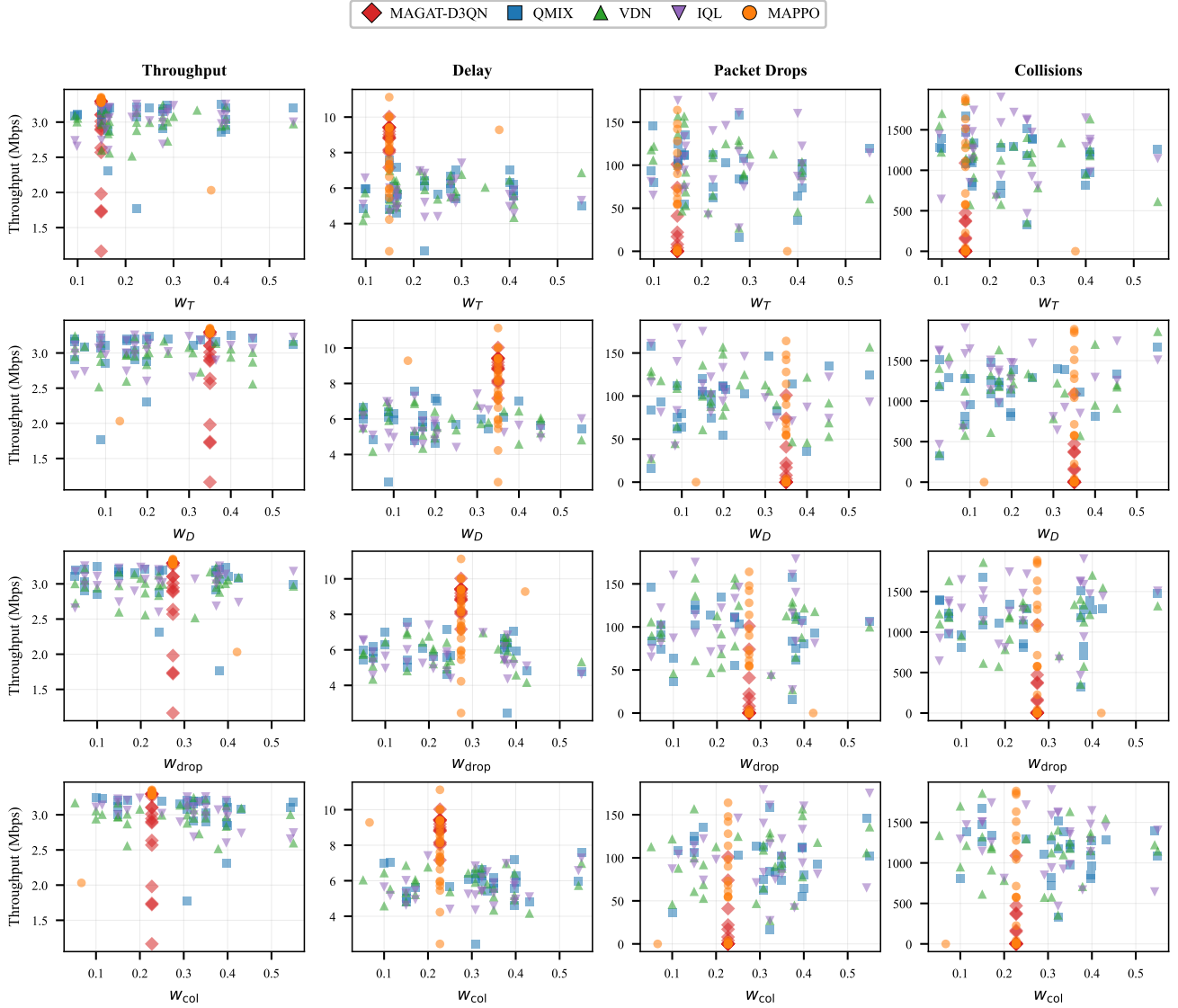


Fig. 7. Reward weight sensitivity analysis. Each row corresponds to one reward weight (w_T , w_D , w_{drop} , w_{col}) and each column to one objective metric. The scatter reveals that throughput is maximized at moderate $w_T \approx 0.15$ – 0.28 and delay improves with higher w_D , while drop and collision weights show diminishing returns beyond $w_{\text{drop}} \approx 0.27$.

defines the decision units, the observation boundaries, and the graph nodes that participate in decentralized coordination. Consequently, the cluster-dynamics sanity study is not auxiliary. It determines whether the structural layer supporting the controller remains interpretable through bounded reassociation, controlled split/merge activity, and stable graph degree evolution. If the realized cluster process is excessively unstable, then even strong aggregate performance would be difficult to interpret.

The same reasoning applies to local leader handover. The handover rule

$$L_k(t^+) = \arg \max_{u \in C_k(t) \setminus \{L_k(t)\}} H_u(t)$$

is included as a robustness extension that preserves local continuity of control when the current leader becomes unreliable. Its significance is therefore operational rather than foundational. A successful local recovery mechanism strengthens

the decentralized interpretation because it avoids immediate reversion to centralized rescue logic. However, any resilience claim must remain tied to the evaluated failure modes, graph conditions, and load regimes.

A broader implication follows. The learned controller and the cluster-management logic should not be interpreted as separable modules with independent guarantees. The performance of the former depends on the structural fidelity of the latter.

D. Generalization and Robustness Implications

The mismatch and imperfect-information studies address a central reviewer concern: whether the controller has learned reusable structure or merely adapted to one simulator regime. Let

$$e_{\text{train}} = (n_{\text{train}}, \mathcal{M}_{\text{train}}, \mathcal{H}_{\text{train}}, \mathcal{T}_{\text{train}})$$

denote the nominal training regime and

$$e_{\text{test}} \neq e_{\text{train}}$$

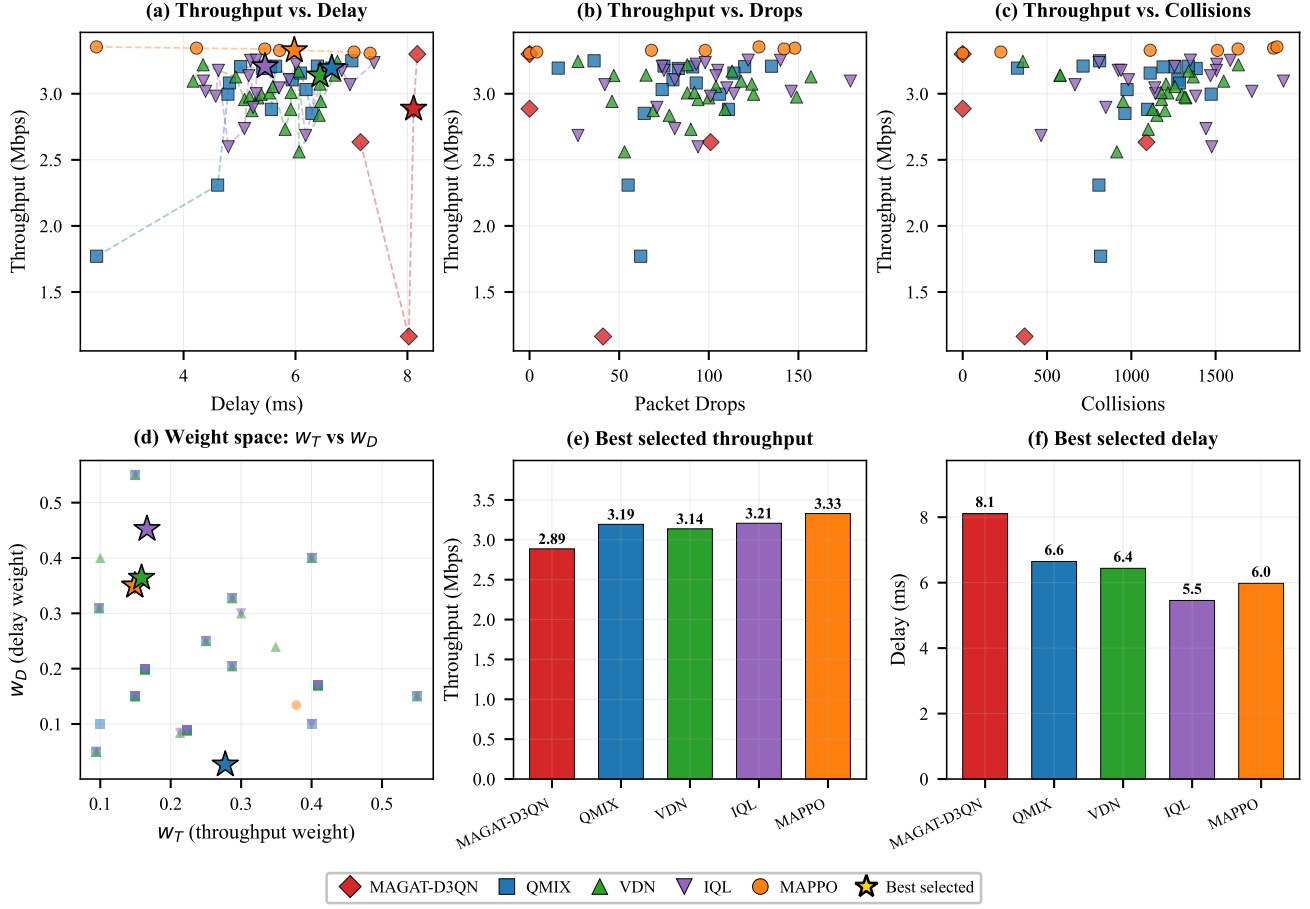


Fig. 8. Combined NSGA-II Pareto analysis: (a–c) Pareto fronts across throughput–delay, throughput–drops, and throughput–collisions trade-off planes with $\star$ marking the best selected (balanced L_2 -to-ideal) configuration per algorithm; (d) reward weight space w_T vs. w_D showing the diversity of explored configurations; (e,f) best selected throughput and delay for each algorithm.

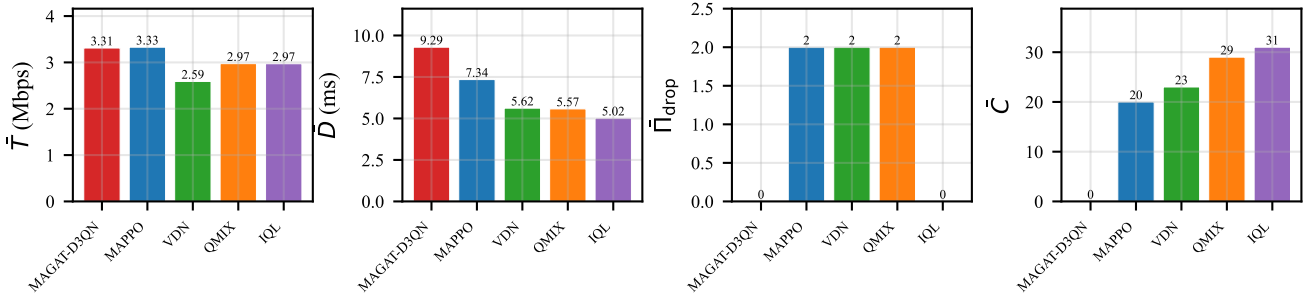


Fig. 9. Main offered-load benchmark: throughput $\bar{T}$, delay $\bar{D}$, drops $\bar{n}_{\text{drop}}$, and collisions $\bar{C}$ across all five MARL algorithms at the nominal evaluation load ($\lambda=525$ pps). MAGAT-D3QN achieves zero drops and zero collisions while maintaining throughput within 1.5% of the best, but incurs the highest mean delay. Lines represent means over 5 evaluation seeds; corresponding mean $\pm$ std values are reported in Table VI.

denote an evaluation regime outside that nominal configuration. The measured relative degradation

$$\Delta_y^{\text{rel}} = \frac{y_{\text{test}} - y_{\text{nominal}}}{|y_{\text{nominal}}|}$$

is therefore the relevant quantity for robustness interpretation, not the absolute value of y_{test} alone.

The same logic applies to imperfect-information studies. Because decentralized execution depends on $\tilde{o}_k(t)$ and $\tilde{G}_t$ rather than a perfect global state, the sensitivity of the controller to stale observations, noisy summaries, or corrupted graph

edges is a first-order property of the architecture. Favorable robustness results would indicate that the graph-aware controller degrades gradually when neighborhood information is perturbed. They would not justify a blanket claim of graph-corruption invariance. Corruption type, severity, and frequency remain essential qualifiers.

Accordingly, the generalization and imperfect-information results should be interpreted as evidence of bounded robustness under the tested regime family, not as universal guarantees.

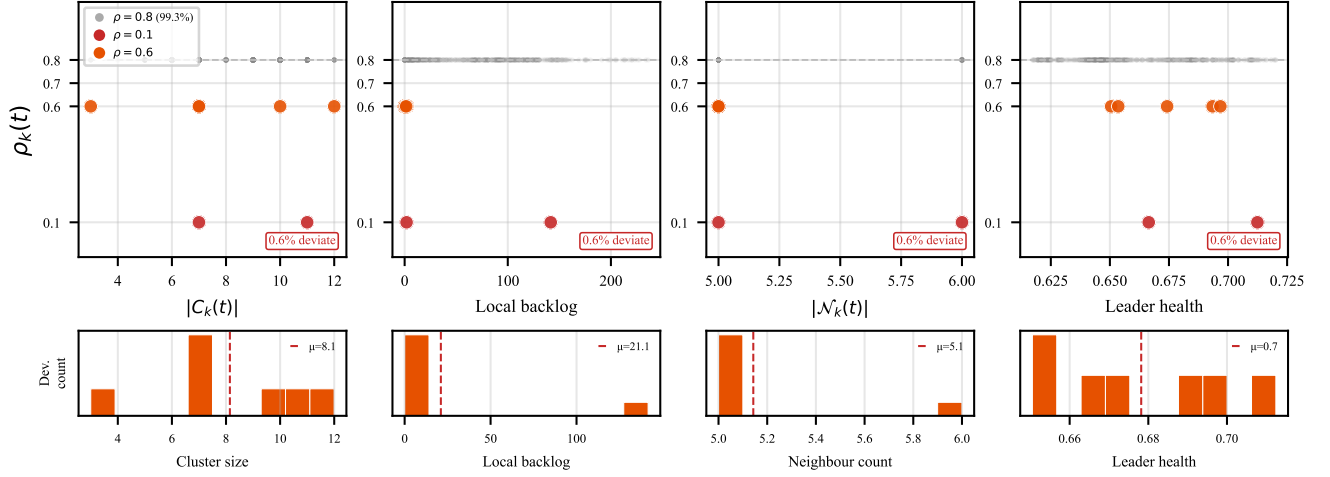
MAGAT-D3QN: $\rho_k(t)$ policy vs. state features (grey = dominant $\rho = 0.8$; coloured = deviation cases)

Fig. 10. Burst-allocation behavior of MAGAT-D3QN: $\rho_k(t)$ plotted against cluster size $|C_k(t)|$, local backlog, neighbor count $|\mathcal{N}_k(t)|$, and leader health. The learned policy concentrates $\rho \approx 0.80$ for intra-cluster service with occasional drops to $\rho=0.10$ for inter-cluster coordination. Points represent means over 5 evaluation seeds; summary statistics are in Table VII.

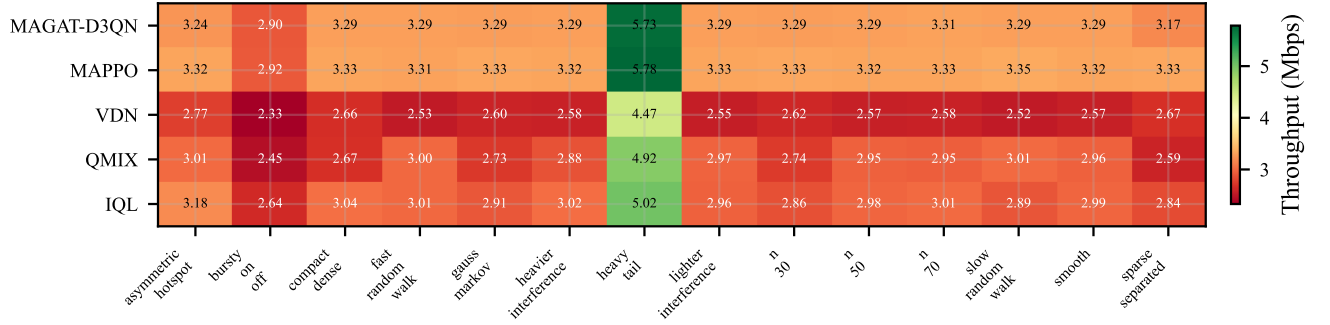


Fig. 11. Generalization under train-test mismatch across 13 scenarios (node count, mobility, interference, traffic, and topology). All methods show $\geq 97\%$ throughput retention under node-count, mobility, and interference mismatch; traffic mismatch produces the largest fairness degradation. Bars represent means over 5 evaluation seeds; mean $\pm$ std values are reported in Table XI.

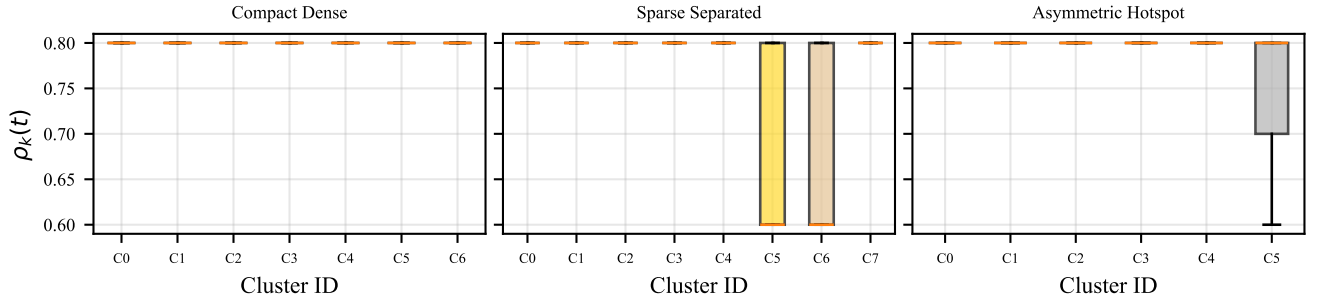


Fig. 12. Cluster-wise $\rho_k(t)$ distributions under three topology configurations: compact-dense, sparse-separated, and asymmetric hotspot. Boxplots show the per-cluster burst allocation patterns, revealing that MAGAT-D3QN adapts ρ based on local cluster topology. Data aggregated over 5 evaluation seeds; summary statistics are in Table VIII.

E. Complexity-Aware Evaluation

The complexity and communication results must be interpreted jointly with the performance gains. A decentralized controller is practically meaningful only if its local computational and messaging burdens remain commensurate with the improvement it provides. In the present architecture, the relevant cost is not only system-wide overhead, but also the burden borne by each cluster-head.

The measured inference quantities

$$\bar{\tau}_{\text{inf}}, \quad \tau_{95}, \quad \tau_{99}$$

and the communication-overhead term

$$B_{\text{comm}}(t) = b_{\text{sum}} c(t) \bar{d}(t)$$

together determine the deployment-facing cost of graph-aware decentralized control. The former reflects decision latency; the

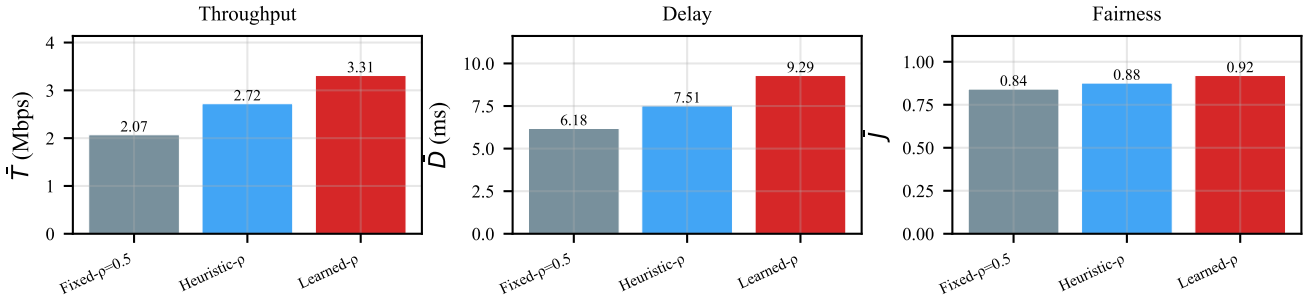


Fig. 13. Fixed, heuristic, and learned burst allocation comparison: throughput $\bar{T}$, delay $\bar{D}$, and Jain fairness $\bar{J}$ at $\lambda=525$ pps. The learned policy achieves 60% higher throughput than fixed- $\rho=0.5$ and 14.5% higher than fixed- $\rho=0.8$, at the cost of increased delay. Bars represent means over 5 evaluation seeds; mean values are in Table VII.

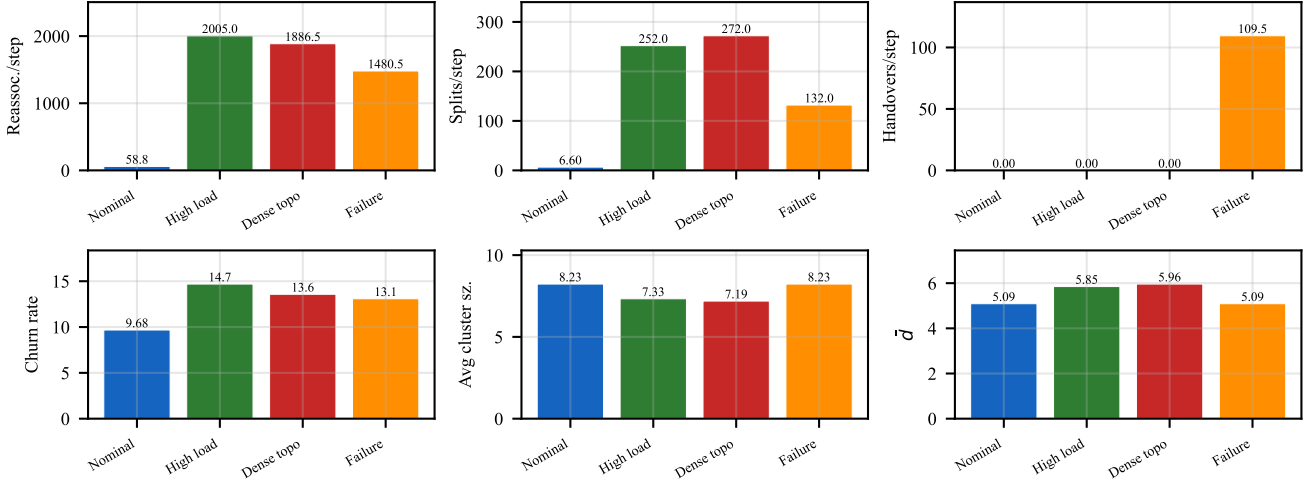


Fig. 14. Cluster-dynamics sanity statistics across four operational scenarios: nominal load, heavy-tail traffic, compact-dense topology, and failure window. Reassociation and split counts are episode totals (cumulative over 100 steps), not per-step rates. Cluster size and graph degree remain stable even under elevated reassociation activity. Data averaged over 5 evaluation seeds; summary in Table IX.

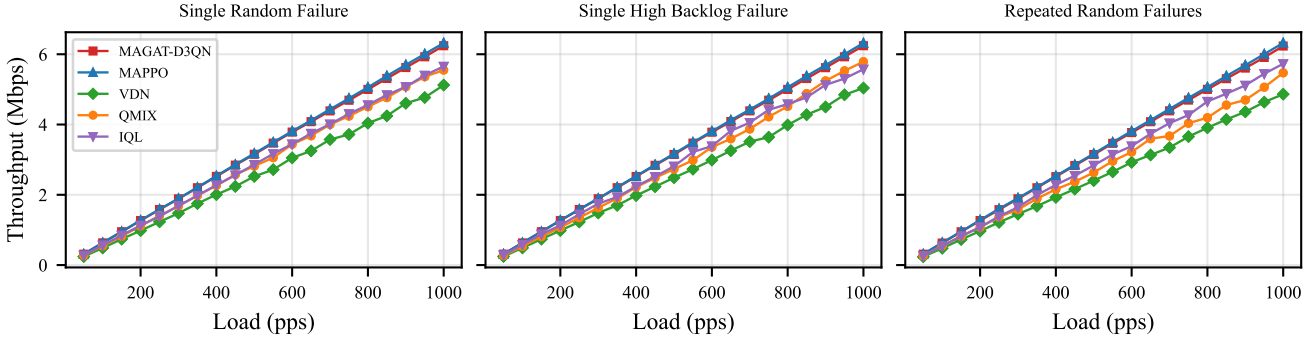


Fig. 15. Leader-failure recovery performance under three failure types: single random failure, single high-backlog failure, and repeated random failures. All five algorithms are evaluated across 20 offered-load levels (50–1,000 pps). Lines represent means over 5 evaluation seeds; mean $\pm$ std values are reported in Table X.

latter reflects the amount of inter-cluster summary exchange induced by coordination. Because both terms can grow with active cluster count and graph density, the relevant deployability question is not whether the method is lightweight in the abstract, but whether the observed cost scaling remains proportionate to the performance and robustness gains obtained in Section VI.

Even strong simulator-side complexity results should therefore be interpreted conservatively. They support software-level plausibility under the measured platform and protocol

assumptions, but they do not by themselves establish readiness for embedded UAV deployment.

F. Limitations and Remaining Open Issues

Several limitations should be stated explicitly.

First, the study remains conditioned on simulator assumptions regarding mobility, channel behavior, synchronization, and traffic generation. Even when the evaluation includes mismatch and imperfect-information stress tests, it cannot represent the full uncertainty of real aerial communication

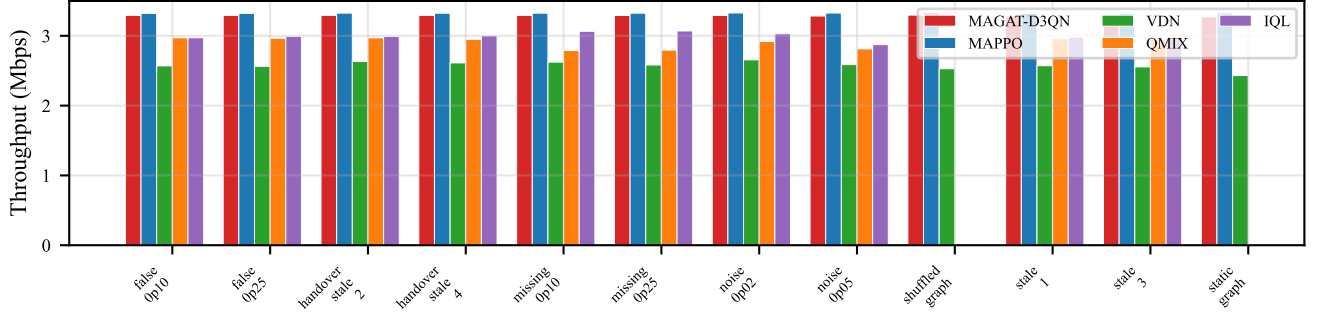


Fig. 16. Robustness to imperfect information: throughput under stale observations, noisy summaries, missing/false graph edges, delayed handover, and static graph conditions. All five algorithms have full coverage across all perturbation types. Bars represent means over 5 evaluation seeds; numerical summary in Table XII.

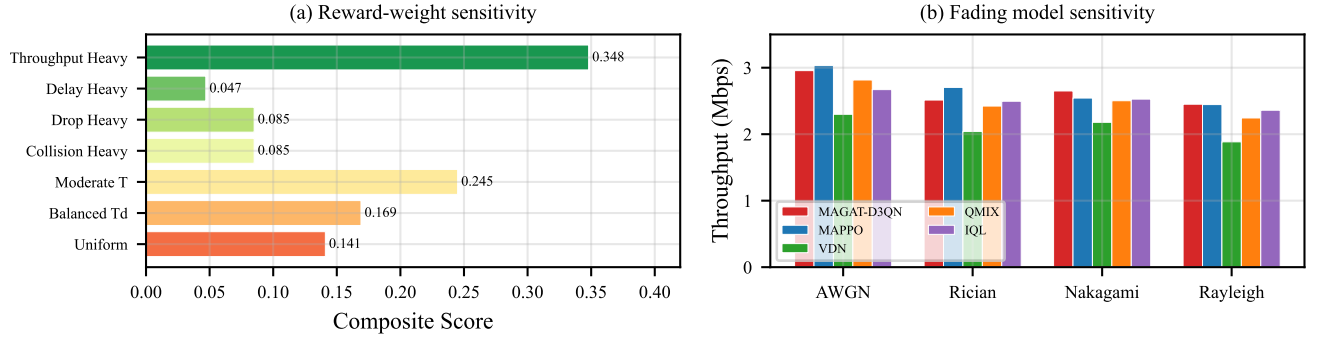


Fig. 17. Environment and reward-weight ablation: (a) composite score across seven reward-weight configurations showing throughput-heavy weighting as optimal, (b) per-algorithm throughput under AWGN, Rician, Nakagami- m ($m=2$), and Rayleigh fading channels. Rayleigh imposes a 17.3% mean throughput reduction; MAGAT-D3QN and MAPPO retain the highest absolute throughput under all fading models. Points represent means over 5 evaluation seeds; summary in Table XV.

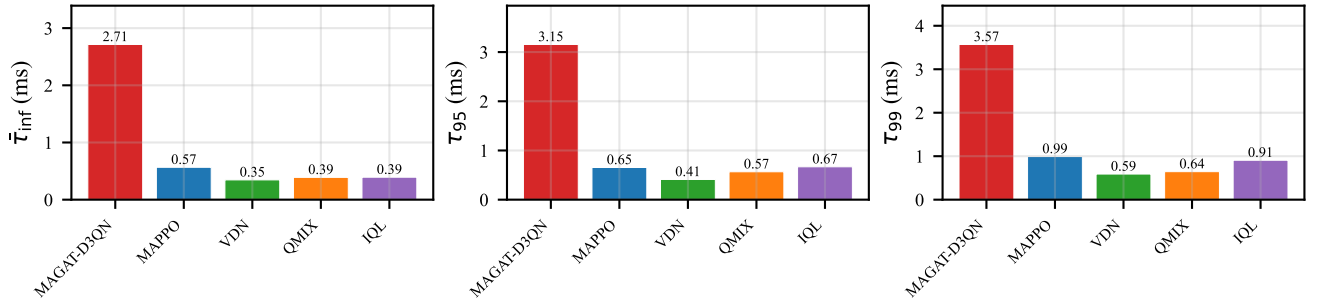


Fig. 18. Complexity and deployability: mean inference latency $\bar{\tau}_{\text{inf}}$, 95th-percentile τ_{95} , and 99th-percentile τ_{99} for all five MARL algorithms. MAGAT-D3QN's graph attention mechanism incurs 5–8 $\times$ higher latency than the simpler baselines. All measurements are under desktop GPU execution and do not represent embedded UAV hardware performance. Summary in Table XVI.

environments, particularly under stronger physical-layer impairments, heterogeneous radios, or asynchronous coordination delays.

Second, the controller depends on the quality of the clustering layer. If cluster formation, reassociation, or local handover logic is poorly tuned, then the resulting observations and graph structure may not represent the true service-coordination problem accurately. In that case, the policy may be limited more by the structural abstraction than by the learning algorithm itself.

Third, the action space is intentionally compact:

$$a_k(t) = (m_k(t), \rho_k(t)).$$

This improves tractability, but it also limits expressiveness.

Extending the control space to finer burst discretization, richer MAC modes, channel selection, or additional cross-layer variables may improve flexibility while also increasing learning difficulty, sample complexity, and coordination overhead.

Fourth, the graph-aware architecture introduces explicit computational and communication costs. These costs are localized rather than centralized, but they are not negligible and may become more pronounced as $c(t)$ and $\bar{d}(t)$ increase. Any observed scaling burden should therefore be reported as a practical limitation rather than treated as an implementation detail.

Finally, the paper does not claim analytical optimality or closed-form performance guarantees for the learned controller.

The contribution is architectural and empirical: a mathematically explicit decentralized clustered FANET formulation, a graph-aware local control policy, and a structured evidence suite for studying service–coordination control under topology variation, imperfect information, and local leader failures.

VIII. CONCLUSION AND FUTURE WORK

This paper studies decentralized control for dynamic clustered FANETs in which cluster-heads act as the decision-making agents over a graph-coupled coordination structure. The problem is formulated at the cluster level with local actions of the form $a_k(t) = (m_k(t), \rho_k(t))$, where $\rho_k(t)$ controls how the burst budget is divided between intra-cluster service and inter-cluster coordination. On that basis, the paper develops a graph-aware MARL controller that operates with local observations and decentralized execution while using a structured centralized SARL baseline only as a comparator.

The evaluation scope is intentionally broader than a single benchmark. In addition to the main offered-load comparison, the paper examines whether the learned burst split behaves meaningfully, whether graph-aware control responds to matched-load topology changes, whether learning $\rho_k(t)$ is preferable to fixed or heuristic burst allocation, whether cluster dynamics remain interpretable, whether local leader handover supports recovery under failures, whether performance degrades gracefully under train-test mismatch and imperfect graph information, and what computational and communication costs accompany decentralized execution. The results show that the graph-aware controller provides zero-drop, zero-collision operation with competitive throughput, but does not achieve the highest raw throughput or the lowest delay among all compared methods. The principal contribution is therefore a better reliability–collision–fairness tradeoff rather than uniform dominance. Taken together, these studies test not only whether the method performs competitively, but also whether the decentralized interpretation remains structurally credible.

Several technical extensions are now directly motivated by the present formulation. A first direction is to increase physical-layer realism through richer air-to-air and air-to-ground channel models, tighter timing uncertainty, and more explicit asynchronous coordination effects. A second direction is to expose cluster-count variation more explicitly as a controlled generalization axis rather than treating it only as an induced property of scenario dynamics. A third direction is to move beyond the current discrete burst-split encoding toward continuous-action burst allocation while preserving stable decentralized training. A fourth direction is to couple the present controller more tightly with routing and relay decisions so that topology-aware burst control and packet forwarding co-adapt rather than being studied in isolation. A fifth direction is to test the architecture under hardware-in-the-loop or small-scale swarm experiments, where communication latency, synchronization error, and onboard compute limits can be measured directly instead of inferred from simulation alone.

These extensions would strengthen the deployment relevance of the present framework, but they do not alter the

central scope of the current paper: a decentralized, graph-coupled cluster-head control formulation for dynamic FANETs together with a disciplined empirical evaluation plan for service-versus-coordination control under changing topology and partial observability.

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